

Endogenous corazonin signaling modulates the post-mating switch in behavior and physiology in females of the brown planthopper and *Drosophila*

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eLife Assessment

This **important** study presents **convincing** evidence that uncovers a novel signaling axis impacting the post-mating response in females of the brown planthopper. The findings open several avenues for testing the molecular and neurobiological mechanisms of mating behavior in insects, although broad concerns remain about the relevance of some claims.

<https://doi.org/10.7554/eLife.109297.1.sa2>

Abstract

Mating in insects commonly induces an alteration in behavior and physiology in the female that ensures optimal offspring. This is referred to as a post-mating response (PMR). The induction of a PMR requires not only male-derived factors transferred with semen during copulation, such as sex peptide (SP) in *Drosophila*, but also intrinsic female signaling components. The latter signaling remains poorly understood in most insects, including the brown planthopper (BPH) *Nilaparvata lugens*, a devastating rice pest. In BPHs the PMR comprises a reduced receptivity to re-mating and increased oviposition. Here, we demonstrate that the neuropeptide corazonin (CRZ) and its receptor (CrzR) are critical for the PMR in female BPHs. Peptide injection and knockdown of CRZ expression by RNAi or CRISPR/Cas9-mediated mutagenesis demonstrate that distensible CRZ signaling suppresses mating receptivity in virgin *N. lugens* females and mediates a reduction in re-mating frequency and increased ovulation. The *CrzR* is highly expressed in the female reproductive tract, and *CrzR*-knockdown phenocopies *Crz* diminishment. Importantly, female *CRZ/CrzR* signaling is indispensable for male seminal fluid factors (e.g. maccessin) to induce the PMR. With disrupted *CrzR* signaling, seminal fluid or maccessin injection fails to reduce female receptivity. Notably, CRZ is not produced in male accessory gland (MAG) and thus not transferred during copulation. However, male *Crz* knockout impairs the PMR in mated females and combining male and female *Crz* knockouts nearly abolished the PMR, demonstrating that CRZ is essential for PMR generation. Transcriptomics of the MAG

indicates that *Crz* knockout affects the expression of numerous seminal fluid protein genes. Finally, we found that also in female *Drosophila melanogaster*, disrupted *Crz* signaling resulted in increased re-mating and reduced oviposition, while CRZ injection suppressed virgin receptivity and increased oviposition. In summary, our study reveals that endogenous female CRZ signaling and male-derived signals cooperate to regulate post-mating transitions in BPHs and *Drosophila*.

Introduction

Reproductive behavior is critical for population sustenance and survival of species, and is influenced by a complex array of internal conditions, environmental factors, and social interactions. In insects, males typically initiate mating through courtship, while females determine acceptance by integrating external sensory stimuli with their internal state. Consequently, female reproductive activity is modulated by multiple signaling pathways ^{1–8}. In insects, successful copulation commonly induces profound physiological and behavioral changes in the females ^{5,9–18}.

Following their first mating, insect females typically reject further courtship attempts and initiate oviposition. This distinct shift in behavior and physiology is referred to as a post-mating response (PMR) ^{5–7,19–23}. A PMR has been documented across diverse insect taxa, including *Drosophila melanogaster*, *Anopheles gambiae*, *Aedes aegypti* and the brown planthopper (BPH), *Nilaparvata lugens* ^{6,18,21,24,25}.

Induction of a PMR commonly requires the transfer of specific seminal fluid components during copulation. In *Drosophila*, seminal fluid peptides, including sex peptide (SP) and DUP99B produced in male accessory glands (MAG), enter the female reproductive tract, bind to specific receptors on sensory neurons, and activate signaling pathways that drive post-mating behavior and physiology ^{6,13,14,18,26}. Additional factors such as Acp26Ab and ovulin (Acp26Aa) are also transferred with the seminal fluid and modulate female reproductive physiology in *Drosophila* ^{27–30}. Notably, it has been shown that the PMR-inducing substances, produced in the MAG of various insects, are highly species-specific and not even conserved across related taxonomic groups ^{25,31}.

While MAG-derived sex peptide (SP) and its receptor (SPR) are key regulators of the female post-mating response (PMR) in *Drosophila* ^{5–7,16}, additional signaling pathways intrinsic to females also modulate this process. Interneurons within abdominal neuromeres of the ventral nerve cord (VNC) promote receptivity to males by signaling via myoinhibitory peptide (MIP) to brain circuits ³². Furthermore, the neuropeptide diuretic hormone 44 (DH44) modulates female sexual receptivity through a sex-specific brain circuit; mating suppresses DH44 signaling, thereby reducing receptivity ¹. During sexual maturation, leucokinin (LK)-expressing neurons (ABLKs) in the abdominal ganglion, downstream of SP-activated circuitry, suppress female receptivity ³³. In addition to changes in receptivity, the *Drosophila* PMR includes a shift in diet preference toward yeast consumption. This altered feeding behavior is mediated by specific peptidergic neurons (ALKs) in the brain through LK signaling ¹².

Although the mechanisms and neuronal pathways underlying the post-mating response (PMR) in *Drosophila* are well characterized ^{3,4,6,7,34–37}, this process remains poorly understood in most non-model insects. A prominent example is the brown planthopper (BPH), *N. lugens* (Hemiptera). As a monophagous rice pest, the BPH exhibits robust reproductive capacity and remarkable environmental adaptability ³⁸. It has also developed high resistance to diverse chemical pesticides ^{39–41}. Consequently, elucidating reproductive physiology and behavior in BPHs is crucial for developing novel control strategies.

In BPHs, the recently identified MAG-derived peptide maccessin (macc) is transferred during copulation and triggers a PMR in females, although its receptor remains unknown²⁵. Notably, the same study demonstrated that a second peptide, ion transport-like peptide 1 (ITPL-1), also modulates the PMR. Whereas macc is male-specific, ITPL-1 is produced both in the MAG and endogenously in females, where it reduces female receptivity to courting males²⁵. Given its endogenous female expression, ITPL-1 likely functions as an intrinsic signaling component in the PMR of BPHs, analogous to the role of myoinhibitory peptide (MIP) and DH44 in *Drosophila* females^{1,32}.

Here, we explored the role of another neuropeptide, corazonin (CRZ), in the reproductive behavior of BPHs. In *Drosophila* males, CRZ is released from interneurons of the ventral nerve cord (VNC) and acts on serotonergic projection neurons to regulate ejaculation and copulation duration⁴². Additionally, CRZ release in males during mating mitigates mating-induced heart rate acceleration, to promote physiological recovery⁴³. CRZ furthermore influences post-mating dietary preference in males¹². Surprisingly, the role of CRZ signaling in female reproduction remains unexplored. To fill this gap, we examined the role of CRZ in female reproduction, including the PMR, in BPH and *Drosophila melanogaster*.

In this study, we demonstrate that CRZ induces a post-mating switch in receptivity in females of both BPH and *Drosophila*. Notably, CRZ injection in virgin females diminishes receptivity to courting males, while *Crz* knockdown or knockout in mated females renders them receptive to further mating. Importantly, CRZ is not produced in the MAG, and thus not transferred from males, but acts as an endogenously produced peptide in females. Furthermore, we detected strong expression of the CRZ receptor (*CrzR*) in the female reproductive tract of BPHs. We demonstrated that both CRISPR-Cas9-mediated *CrzR* knockout and RNAi-mediated *CrzR* knockdown disrupt the PMR, phenocopying *Crz* knockdown. Critically, we could show that the level of receptivity to mating in virgin females, whether by MAG extract (with seminal fluid proteins) injection or macc injection, is strictly dependent on presence of the *CrzR* receptor. Importantly, manipulations of CRZ signaling in female *Drosophila* yield similar effects on receptivity and oviposition. Interestingly, we find that knockdown or knockout of *Crz* in male *N. lugens* does not affect male copulation behavior, but *Crz* deficient males do not induce a full PMR in mated females. Notably, knockout of *Crz* in both male and female partners results in an almost complete abolishment of the PMR. Since CRZ is not produced in the MAG we performed a transcriptomic analysis of the MAG in *Crz* mutants to screen for other candidate CRZ-regulated genes affecting the PMR. We found altered expression of a large number of genes, containing seminal protein genes, but no change in the genes encoding *macc* and *ITPL-1*, suggesting presence of other MAG factors, yet to be identified. In summary, our findings strengthen the notion that the female PMR requires not only transfer of male-derived factors, but also endogenous female neuropeptide signaling components.

Results

CRZ signaling reduces receptivity to courting males and stimulates oviposition in *N. lugens* females

While numerous studies have established roles of the neuropeptide CRZ in regulating aspects of male reproduction across diverse insect species^{44–50}, research addressing its potential functions in female reproduction remains conspicuously lacking.

This study investigates the role of CRZ in female reproduction, starting with the brown planthopper (BPH), *Nilaparvata lugens*, a major agricultural pest. Initially, the *Crz* gene (GenBank accession: AB817247.1) was cloned based on transcriptome data⁵¹. The deduced mature CRZ sequence in BPH, pQTFQYSRGWTamide, is identical to that reported for most studied insect species (Figure 1-figure supplement 1A). Consistent with other known CRZ precursors, the BPH

precursor encodes a single CRZ peptide starting at the first amino acid of the propeptide (**Figure 1-figure supplement 1A** [↗](#) and B). A comparison of the organization of selected CRZ precursor genes is shown in **Figure 1-figure supplement 1B** [↗](#). Genomic analysis revealed that the CRZ precursor is encoded by two exons separated by a single intron (**Figure 1-figure supplement 1B** [↗](#)). The mature CRZ peptide and a scrambled control peptide (sCRZ) were synthesized for pharmacological experiments.

First, we investigated the role of CRZ signaling in the reproductive behavior and PMR of female BPHs. The PMR in BPHs, is characterized by reduced receptivity to courting males, including display of rejection behaviors, and increased oviposition ²⁵[↗](#). This response is partially induced by the male-derived peptide maccessin (macc), transferred during copulation ²⁵[↗](#). We found that injection of CRZ significantly reduced receptivity of virgin female BPHs to courting males (**Figure 1A** [↗](#) and B). Furthermore, virgin females injected with CRZ actively rejected courting males, exhibiting a distinct rejection behavior characterized by ovipositor extrusion and kicking or fleeing. This behavior was not observed in control females injected with sCRZ (**Figure 1 - Supplementary Video 1** [↗](#)). Notably, the CRZ-induced mating refusal behavior in virgin females was no longer observed 24 hours after injection (**Figure 1-figure supplement 2A** [↗](#)). While injection of CRZ peptide did not stimulate egg production in virgin BPHs (**Figure 1-figure supplement 2B** [↗](#)), it significantly promoted oviposition in mated females (**Figure 1C** [↗](#)).

To further investigate CRZ action, we performed RNA interference (RNAi) by injecting double-stranded *Crz* RNA (*dsNlcrz*) into female BPHs, which efficiently reduced *Crz* expression (**Figure 1-figure supplement 2** [↗](#)). Knockdown of *Crz* resulted in increased receptivity with 10% of the mated females mating again, and significantly reduced oviposition to approximately 60% of control levels (**Figure 1E** [↗](#) and F). In contrast, *Crz* knockdown had no effect on the receptivity of virgin females (**Figure 1E** [↗](#)).

Since RNAi-mediated knockdown of *Crz* in female BPHs left a residual level of gene expression (**Figure 1-figure supplement 2** [↗](#)), we generated two *Crz* deletion alleles, ΔCrz^1 and ΔCrz^2 , using CRISPR/Cas9 (**Figure 1G-G** [↗](#)) to achieve a complete ablation. Both alleles harbor deletions eliminating the mature CRZ peptide coding sequence (**Figure 1G** [↗](#)). Sanger sequencing confirmed that the deletions compromised essential triplet codons within the *Crz* open reading frame (**Figure 1-figure supplement 4** [↗](#)), thereby efficiently disrupting CRZ peptide expression (**Figure 1G** [↗](#) and H). Immunohistochemical analysis further validated the mutants, revealing a complete absence of anti-CRZ immunolabeling in homozygous ΔCrz^1 and ΔCrz^2 females (**Figure 1H** [↗](#)). Homozygous ΔCrz^1 and ΔCrz^2 virgin females are fully viable and fertile. They exhibit a courtship vigor similar to controls and their initial mating success rate with wild-type males exceeds 60% (**Figure 1I and J** [↗](#)). However, 25% of the mated mutant females were receptive to further mating (**Figure 1J** [↗](#)), and oviposition was significantly reduced to approximately 50% of the wild-type levels (**Figure 1K** [↗](#)). Collectively, these results demonstrate that diminished CRZ signaling severely perturbs the PMR in female BPHs, seen as increased post-mating receptivity and reduced egg-laying. Importantly, the *Crz* deletion abolished the onset of typical post-mating rejection behavior. We, thus, conclude that endogenous CRZ signaling is important for the induction of a PMR in female BPHs.

CRZ is expressed in similar neurons in both sexes of BPHs and is absent from the male accessory gland

To further understand how CRZ regulates the PMR in BPHs, we examined the expression of CRZ and *Crz* across different tissues in males and females. The *Crz* mRNA expression was examined in various BPH tissues using both RT-PCR and qPCR (**Figure 2A** [↗](#)). We detected *Crz* mRNA expression primarily in the central nervous system (CNS), regardless of the primer set or detection method used, while expression was nearly undetectable in other tissues, including reproductive organs of both sexes (**Figure 2B and C** [↗](#)).

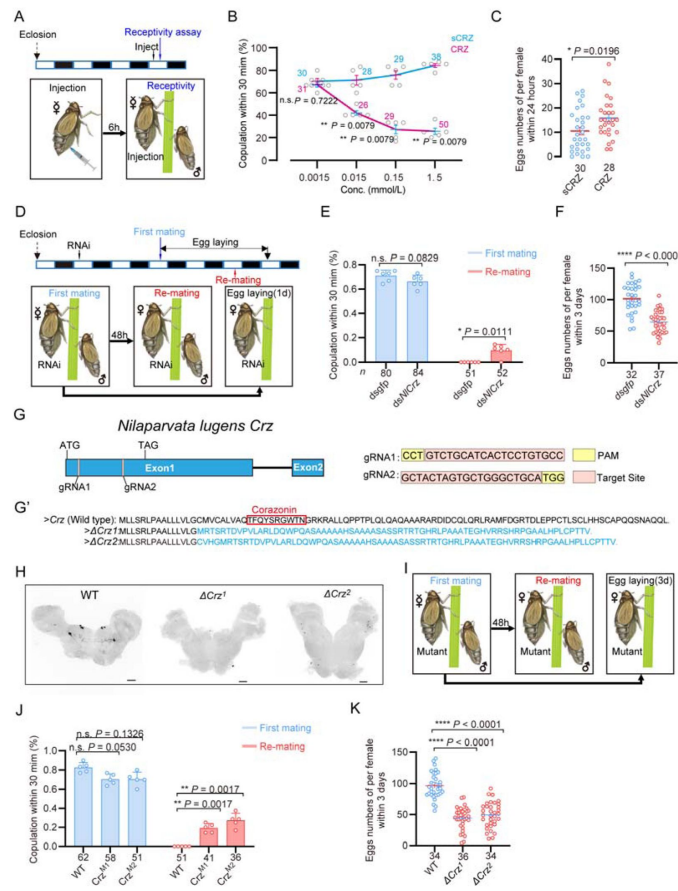


Figure 1.

CRZ signaling affects the post-mating response (PMR) and oviposition in female *N. lugens*.

Wild-type males were used in all behavioral assays. (A) Experimental design for panels B-C. The white/black segments on the experimental time lines denote light/dark periods, respectively. (B) Receptivity of virgin females 6 h post-injection, using different doses of CRZ (sCRZ as control), each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 100 ng/10 ng/1 ng/0.1 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. The graph shows percentage females copulating within 30 min. ** $P < 0.01$ vs. control; ns (non-significant): $P > 0.05$ (Mann-Whitney test). The numbers above the curves denote total number of animals. (C) Eggs laid per mated female 12 h post-copulation. We used the highest CRZ dose from panel B (sCRZ, control). Each mated female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 100 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. * $P < 0.05$ (Student's *t*-test). Data: mean ± s.e.m. (≥3 biological replicates, ≥8 insects/replicate). The numbers below the bars denote total number of animals. (D) Experimental design for panels E-F. (E) Receptivity of virgin (1st mating) and mated females after *Crz*-RNAi (*dsgfp* as control). Graph displays the percentage of females copulating within 30 min. The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean ± s.e.m. * $P < 0.05$, and ns (non-significant), $P > 0.05$, two-way ANOVA followed by Holm-Šidák's multiple comparisons test. (F) Total numbers of eggs laid per female over 72 h after *Crz*-RNAi (*dsgfp* as control). **** $P < 0.0001$ (Student's *t*-test). The small circles and the numbers below the bars denote total number of animals. Data are shown as mean ± s.e.m. (≥3 biological replicates, ≥8 insects/replicate). (G) Schematic diagram of the *Crz* gene and sgRNA (single-guide RNA) design for *Crz* mutant production. The *Crz* gene consists of 2 exons, and ATG and TAG are located on Exon 1. The target site of the 20 bp sgRNA on Exon 1 is highlighted in pink. The PAM sites are indicated in yellow. The two knockout strains ΔCrz^1 and ΔCrz^2 were unable to produce mature CRZ peptides. (G') The amino acid sequences obtained by translating the mutated base sequences. (H) Absence of CRZ immunoreactivity in homozygous ΔCrz^1 and ΔCrz^2 mutants compared to wild type (WT) expression. Scale bars: 50 μ m. (I) Experimental design for panels J-K. (J) Receptivity of virgin/mated females across genotypes. No effect was seen on virgin receptivity, only the re-mating was affected. The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean ± s.e.m. ** $P < 0.01$; *** $P < 0.001$, and ns (non-significant), $P > 0.05$, two-way ANOVA followed by Holm-Šidák's multiple comparisons test. (K) Eggs laid 72 h post-mating. The numbers below the bars denote total number of animals. Data are shown as mean ± s.e.m. **** $P < 0.0001$; Student's *t* test.

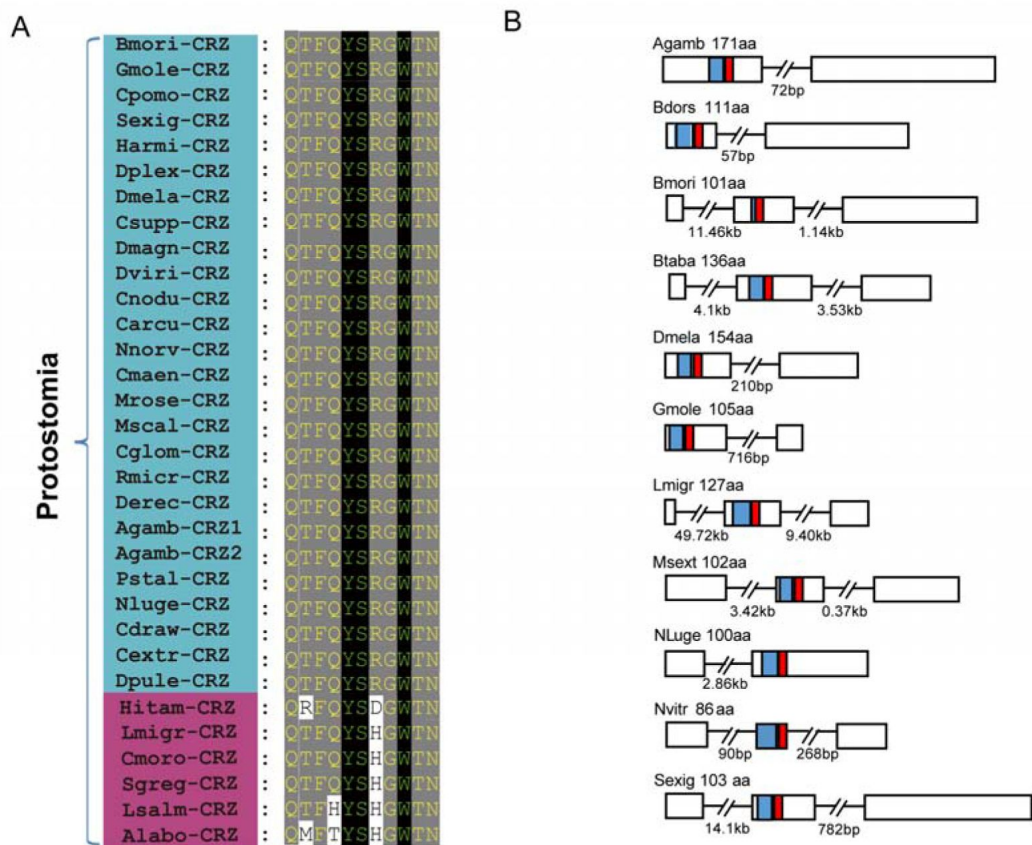


Figure 1 figure supplement 1.

Evolutionary conservation of CRZ peptide sequences in *Nilaparvata lugens* and other arthropods.

(A) Alignment of mature CRZ peptide sequences across species. Abbreviations of species names are as follows: Bmori (*Bombyx mori*), Gmole (*Grapholita molesta*), Cpomo (*Cydia pomonella*), Sexig (*Spodoptera exigua*), Harmi (*Helicoverpa armigera*), Dplex (*Danaus plexippus*), Dmela (*Drosophila melanogaster*), Csupp (*Chilo suppressalis*), Dmagn (*Daphnia magan*), Dviri (*Drosophila virilis*), Cnodu (*Catagly nodus*), Carcu (*Callinectes arcuatus*), Nnorv (*Nephrops norvegicus*), Cmaen (*Carcinus maenas*), Mrose (*Macrobrachium rosenbergii*), Mscal (*Megaselia scalaris*), Cglom (*Cotesia glomerata*), Rmicr (*Rhipicephalus microplus*), Derec (*Drosophila erecta*), Agamb (*Anopheles gambiae*), Pstal (*Plautia stali*), Nluge (*Nilaparvata lugens*), Cdraw (*Caerostris darvini*), Cextr (*Caerostris extrusa*), Dpule (*Daphnia pulex*), Hitam (*Heterotrigona itama*), Lmigr (*Locusta migratoria*), Cmoro (*Carausius morosus*), Sgreg (*Schistocerca gregaria*), Lsalm (*Lepeophtheirus salmonis*), Alabo (*Apis laboriosa*). Species within the blue area possess highly conserved mature peptide sequences, whereas those in the magenta zone exhibit relatively less conservation in their mature peptides. (B) Schematic organization of CRZ precursors in representative species. Signal peptides (blue) and mature CRZ peptides (red) are indicated.

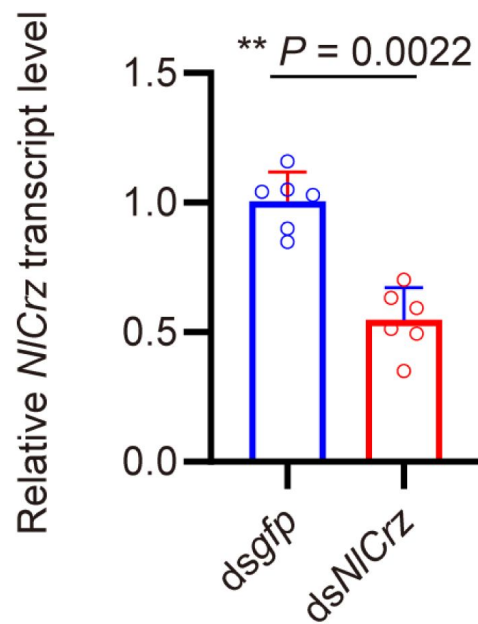


Figure 1 figure supplement 2.

The *Crz* gene-silencing efficacy in female insects following dsRNA injection assayed by qPCR.

The small circles denote the number of replicates. Data are shown as mean ± s.e.m. Mann-Whitney test. *** $P < 0.001$.

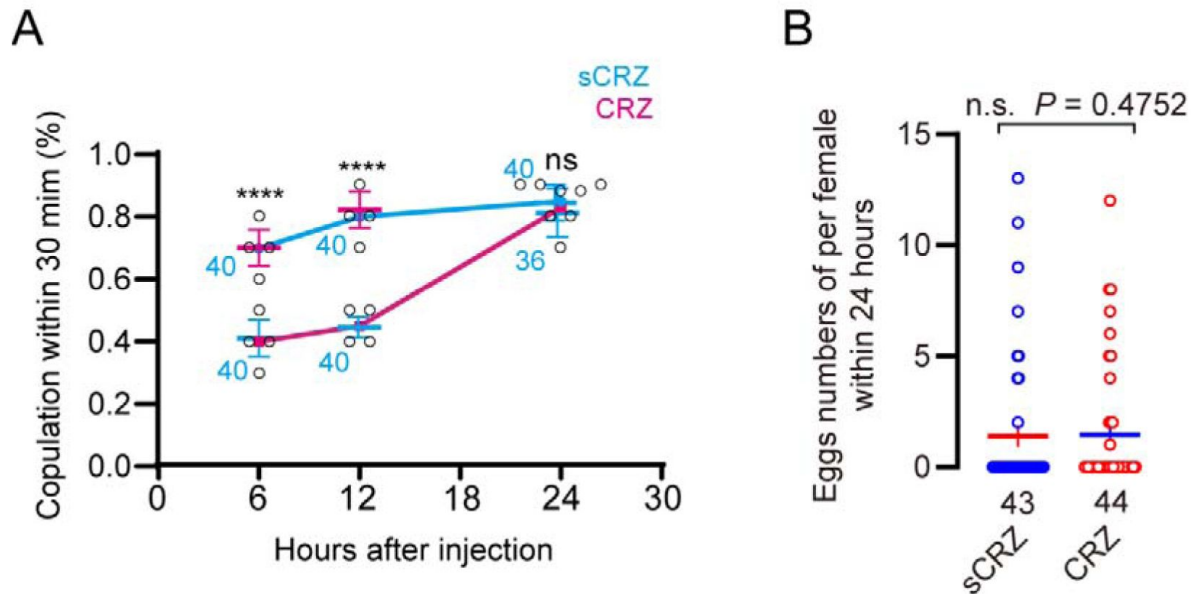


Figure 1 figure supplement 3.

Phenotypic effects of CRZ peptide injection and *Crz* gene knockout.

(A) Receptivity of virgin females at indicated time points after CRZ injection (sCRZ as control). Each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 100 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. Graph shows percentage females copulating within 30 min. **** $P < 0.0001$; ns: not significant (Chi-square test). The numbers associated with the curves denote total number of animals. (B) Eggs laid per virgin female after neuropeptide injection (sCRZ, control). Females were mated with wild-type males for 12 h prior to assessment. Each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 100 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. Data are shown as mean ± s.e.m. ns: $P > 0.05$ (Student's t-test). The numbers below the bars denote total number of animals. We used at least four biological replicates with at least ten insects per replicate for each experiment.

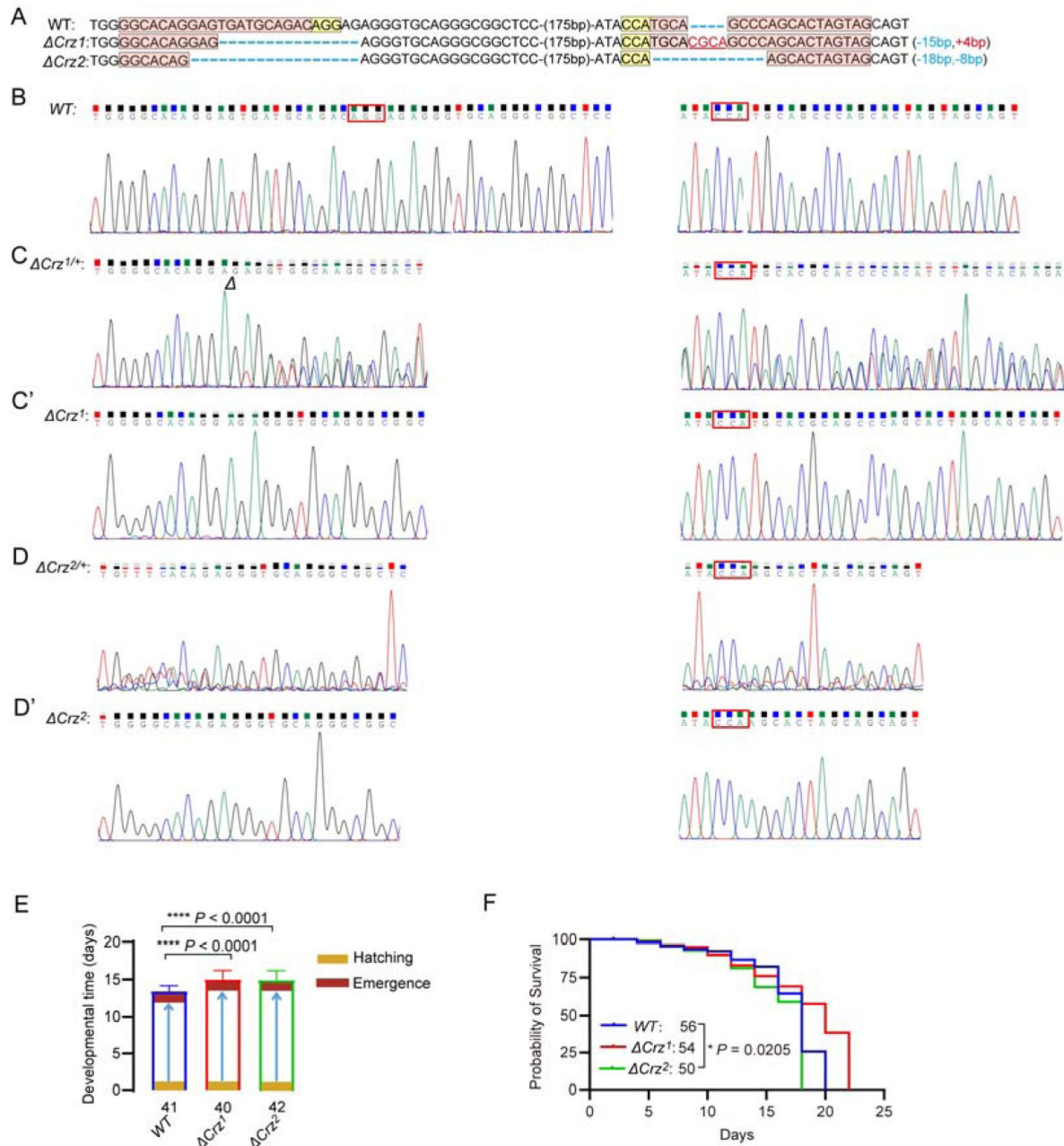


Figure 1 figure supplement 4.

Characterization of *Crz* mutants in *N. lugens*.

(A) Sequencing of *Crz* mutant (ΔCrz^1 and ΔCrz^2) and wild type (WT) PCR products from BPHs. The exact indel types of the G0 mutant individuals were confirmed by cloning and sequencing. The wild-type sequences are shown at the top with the target site marked by in pink shading and PAM in yellow shading. The change in the length of the mutant sequence is shown on the right side of the sequence (+, insertion; -, deletion). (B) The Sanger sequence chromatograms flanking the sgRNA target site for wild type (WT). (C and D) The Sanger sequence chromatograms flanking the sgRNA target site for wild type (WT), heterozygous mutant ($\Delta Crz^{1/+}$ and $\Delta Crz^{2/+}$). (C' and D') The Sanger sequence chromatograms flanking the sgRNA target site for wild type (WT), homozygous mutant (ΔCrz^1 and ΔCrz^2). (E) Developmental duration (egg to adult) across genotypes. **** $P < 0.0001$ (Student's *t*-test). The numbers below the bars denote total number of animals. (F) Survival curves of adult *Crz* mutants vs. WT. * $P < 0.05$ (Log-rank Mantel-Cox test; *df* = 1; *n* $\geq$ 50/group).

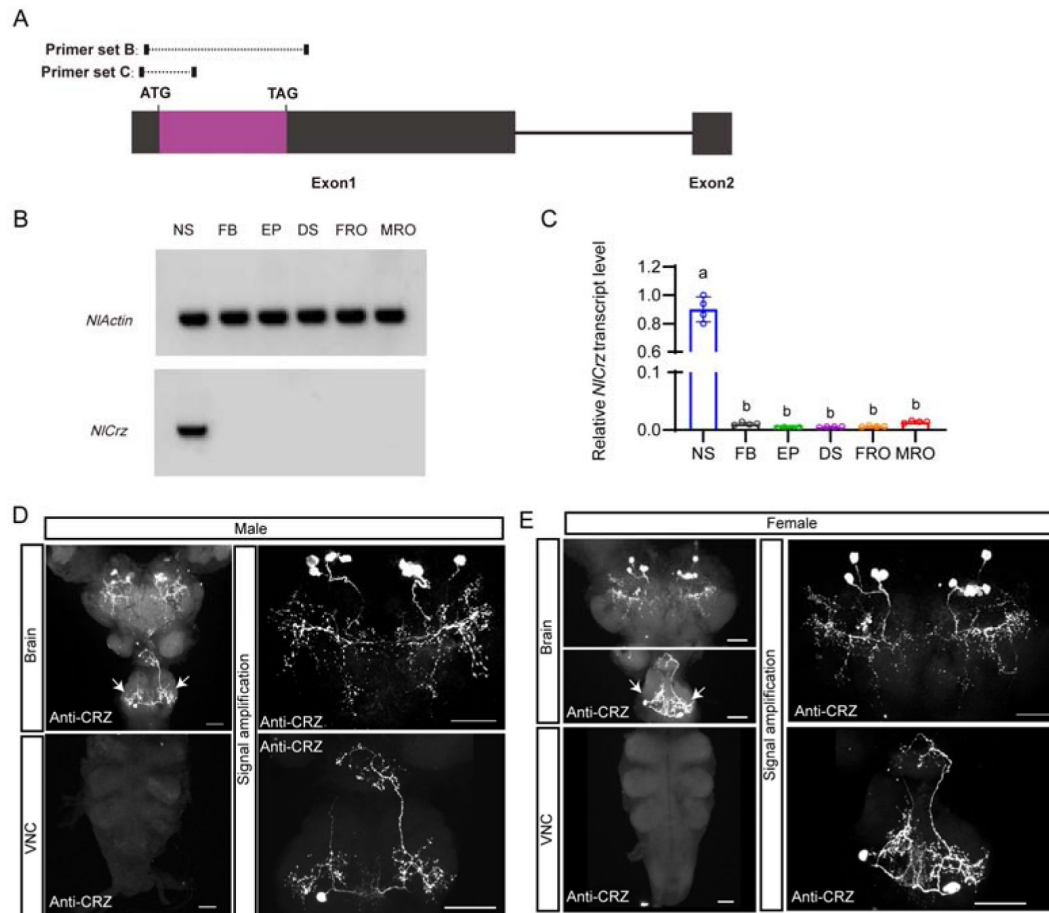


Figure 2.

Expression of *Crz* and CRZ in the nervous system of the brown planthopper (BPH).

(A) Schematic of primer design for *Crz* detection (primer sequences provided in Table S1). (B) RT-PCR analysis of *Crz* mRNA levels in various BPH tissues. Actin served as the loading control. Tissues: Nervous system (NS), fat body (FB), Epidermis (EP), digestive system (DS), female reproductive organs (FRO), male reproductive organs (MRO). Note: Tissues except reproductive organs were pooled from both sexes. (C) qPCR analysis of *Crz* mRNA levels in BPH tissues (abbreviations as in B). (D and E) Immunolocalization of CRZ peptide in the nervous system of male (D) and female (E) BPHs. Four pairs of CRZ neurons in the brain and one pair in the subesophageal ganglion are shown. Note that some of the CRZ neurons are likely to be neurosecretory cells with terminations in the corpora cardiaca, others may be interneurons. The extensive branches within the brain suggest CRZ signaling in brain circuits. VNC, ventral nerve cord. Scale bars: 50 μ m. (E) CRZ immunolabeling in the nervous system of female BPHs. Scale bar: 50 μ m.

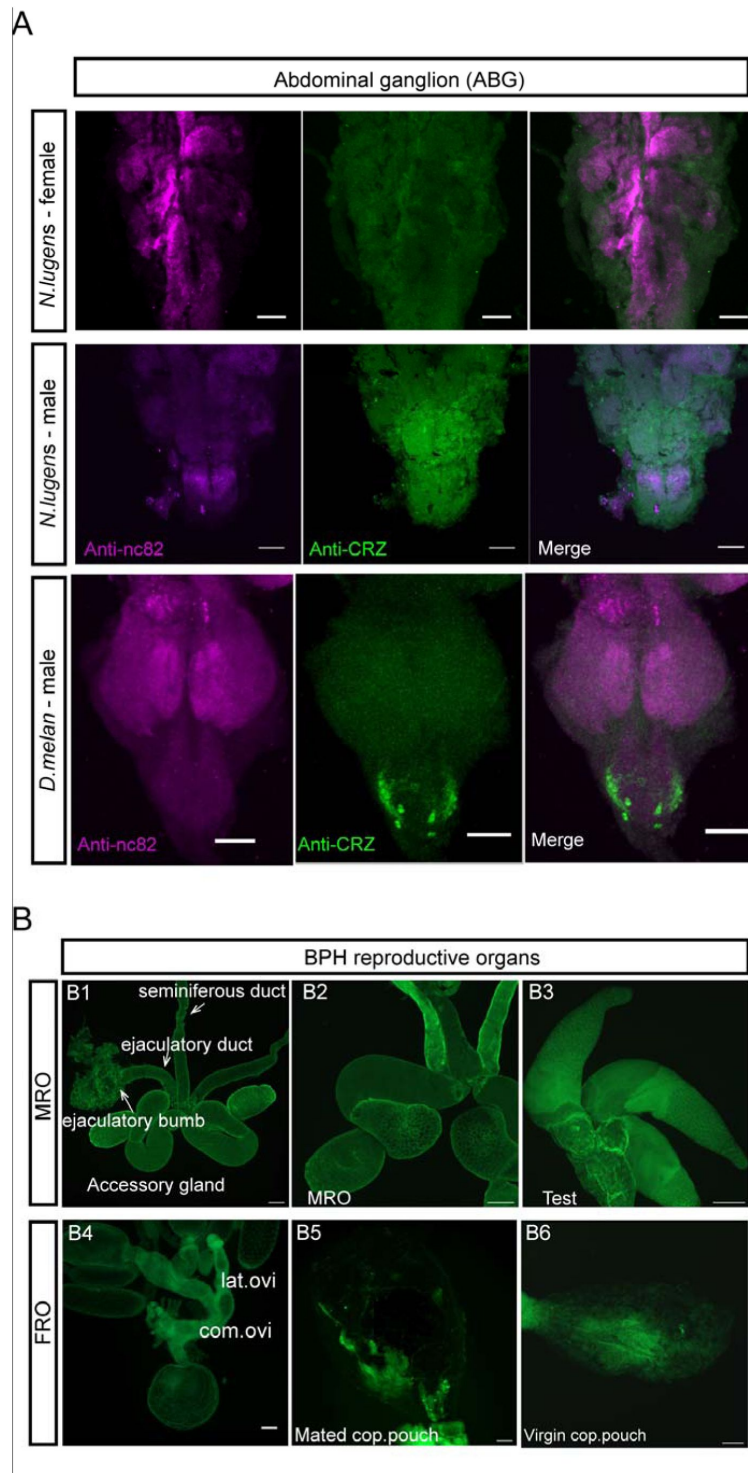


Figure 2 figure supplement 1.

CRZ immunolabeling in abdominal ganglion and reproductive organs of male and female BPHs.

(A) Anti-CRZ immunolabeling in the abdominal ganglia of the ventral nerve cord (VNC) in *N. lugens* and *Drosophila melanogaster* (*D. Melan.*). Scale bar: 50 μ m. Note: Male-specific CRZ-positive neurons are present in *D. melanogaster*, as previously reported. (B) Lack of CRZ immunoreactivity in BPH reproductive organs. (B1-B3) Male reproductive organs (MRO): testes (test). (B4-B6) Female reproductive organs (FRO): lateral oviduct (Lat. ov), common oviduct (Com. ov). Scale bars: 100 μ m.

A previous study reported CRZ expression in four male-specific interneurons within the abdominal ganglia of *Drosophila*⁴⁴. To determine whether CRZ exhibits sexually dimorphic expression in BPHs, we performed immunolabeling using an anti-CRZ antiserum. This labeled four pairs of neurons in the protocerebrum of the brain (**Figure 2D and E**) and one pair of neurons in the subesophageal ganglion in both males and females (**Figure 2D and E**). Since CRZ immunolabeled neurons in brains of most, if not all, insects studied include three or more pairs of lateral neurosecretory cells^{52,53}, we suggest that CRZ signaling regulating the PMR may be hormonal (at least in part). Importantly, no CRZ-immunopositive neurons were detected in the abdominal ganglia of either sex (**Figure 2D and E**), and thus there is no evidence for sex-specific expression of CRZ neurons in BPHs. This is in contrast to *Drosophila*, where we confirmed, as previously reported⁴², the presence of CRZ-immunolabeled neurons specifically in the abdominal ganglia of males (**Figure 2 - figure supplement 1A**).

To exclude that CRZ acts as a seminal fluid peptide transferred to females during copulation, similar to sex peptide (SP) in *Drosophila* and macc in BPHs²⁵, we specifically examined CRZ expression in reproductive tissues of both sexes. However, we observed no CRZ immunolabeling in the male reproductive organs (**Figure 2 - figure supplement 1B1 - B3**) or female reproductive organs (**Figure 2 - figure supplement 1B4 - B6**). This aligns with our transcriptional *Crz* data shown above (**Figure 2B and C**). Furthermore, *Crz* mRNA and CRZ protein were absent from transcriptomic and proteomic datasets of the male accessory glands (MAGs) in BPHs²⁵. Collectively the available data suggest, that CRZ is not produced by the MAG and is consequently unlikely to be transferred to females as a seminal fluid component.

The CRZ receptor is essential for mediating the post mating response in female BPHs

Given our findings implicating CRZ in modulating the PMR in female BPHs, we identified the CRZ receptor (*CrzR*) gene in this species; it encodes a structurally conserved GPCR orthologous to other insect *CrzRs* (**Figure 3-supplemental figure 1**). We next explored the effect of knocking down the *CrzR* on the PMR in BPHs. RNAi-mediated knockdown of the receptor (*dsCrzR* injection) resulted in 15% of the mated females re-mating, a phenotype not seen in *dsgfp*-injected controls (**Figure 3A and B**) and in a significantly reduced oviposition (**Figure 3C**). Importantly, CRZ injection failed to diminish receptivity in *CrzR*-knockdown virgins (**Figure 3D and E**). Furthermore, MAG extract (with seminal fluid proteins) injection into *dsCrzR*-treated virgins did not affect receptivity (**Figure 3F**).

To further confirm the role of the *CrzR* in the PMR of BPHs, we applied CRISPR/Cas9 genome editing to mutate the *CrzR* gene and constructed a homozygous mutant strain of *CrzR* (*CrzR^M*) by genetic crosses (**Figure 3G-3G**, **Fig. 3 - figure supplement 2A**). We found that the developmental time and life span of the female *CrzR* mutants, like the *Crz* mutants, were slightly extended compared to wild type animals (**Figure 3** - supplement figure 2D and 2E). Homozygous *CrzR^M* females displayed increased remating rates (30%) (**Figure 3I and J**) and the number of eggs laid by *CrzR^M* females was also significantly decreased (**Figure 3K**). Consistent with the RNAi results, neither CRZ nor MAG extract injections reduced receptivity in *CrzR^M* virgins (**Figure 3K - M**). Finally, macc peptide injection also failed to decrease receptivity in *CrzR*-mutant virgins (70% acceptance rate), whereas it significantly suppressed mating in wild-type controls (**Figure 3N**). These results collectively demonstrate that intact CRZ/*CrzR* signaling in female BPHs is required for transducing the PMR-inducing effects of male-derived seminal factors, including macc²⁵.

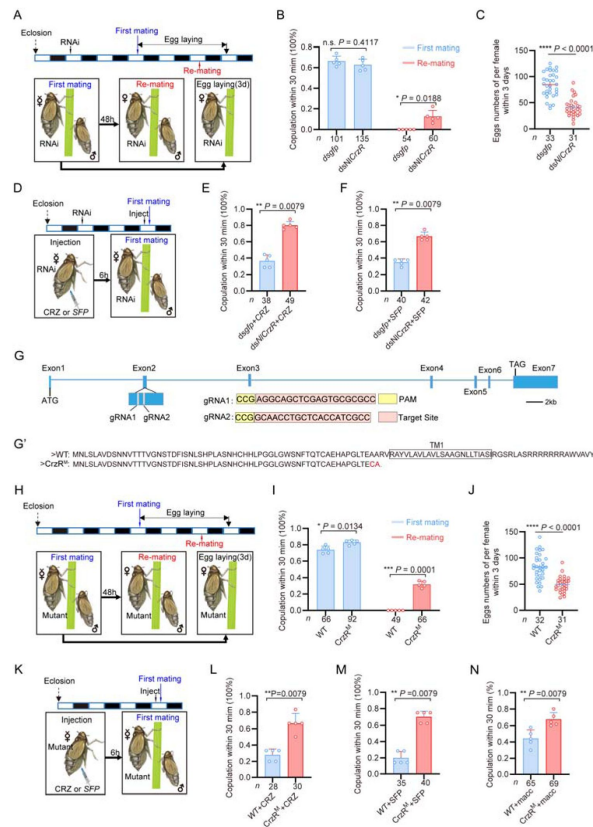


Figure 3.

The CRZ receptor (*CrzR*) is essential for the female PMR and is required for action of MAG-derived factors.

(A) Experimental design for B and C. (B) Receptivity after *CrzR*-RNAi (injection of ds*CrzR*; ds*GFP* as control). Percentage females copulating within 30 min. The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean $\pm$ s.e.m. * $P < 0.05$, and ns (non-significant), $P > 0.05$, two-way ANOVA followed by Holm-Šidák's multiple comparisons test. (C) Numbers of eggs laid within 72 h post-mating (ds*CrzR* females $\times$ WT males). Data are shown as mean $\pm$ s.e.m. **** $P < 0.0001$ (Student's *t*-test). The numbers below the bars denote total number of animals. (D) Experimental design for E-F. (E) Receptivity after CRZ injection in *CrzR*-knockdown virgins (ds*GFP* + CRZ control). Graph shows percentage females copulating within 30 min. Each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 10 ng of peptide dissolved in 50 nL of 1 $\times$ PBS balanced salt solution. The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean $\pm$ s.e.m. ** $P < 0.01$ (Mann-Whitney test). (F) Receptivity after MAG extract (with seminal fluid proteins) injection in *CrzR*-knockdown virgins (ds*GFP* + SFP control). Percentage copulating within 30 min. The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean $\pm$ s.e.m. ** $P < 0.01$ (Mann-Whitney test). (G) *CrzR* targeting strategy for generation of mutants. *Top*: Genomic structure (ATG/TAG in Exons 2/7). sgRNA target (pink), PAM (yellow). *Bottom*: CRISPR alleles. (G') Translated receptor mutant sequences. (H) Experimental design for I-J. (I) Receptivity of *CrzR* mutants compared to wild type animals (WT). The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean $\pm$ s.e.m. * $P < 0.05$, *** $P < 0.001$, two-way ANOVA followed by Holm-Šidák's multiple comparisons test. (J) Number of eggs laid by mutants mated with WT males within 72 h. The numbers below the bars denote total number of animals. Data are shown as mean $\pm$ s.e.m. **** $P < 0.0001$; Student's *t* test. (K) Experimental design for L-M. (L) Receptivity after CRZ injection in *CrzR* mutants compared to WT. Each virgin female BPH was injected with a calibrated glass capillary needle directly into the abdomen with 10 ng of peptide dissolved in 50 nL of 1 $\times$ PBS balanced salt solution. The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean $\pm$ s.e.m. ** $P < 0.01$, Mann-Whitney test. (M) Receptivity after SFP injection in *CrzR* mutants compared to WT. The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean $\pm$ s.e.m. ** $P < 0.01$, Mann-Whitney test. (N) Receptivity after macc injection in *CrzR* mutants compared to WT. The small circles denote the number of replicates; the numbers below the bars denote total number of animals. Data are shown as mean $\pm$ s.e.m. ** $P < 0.01$, Mann-Whitney test.

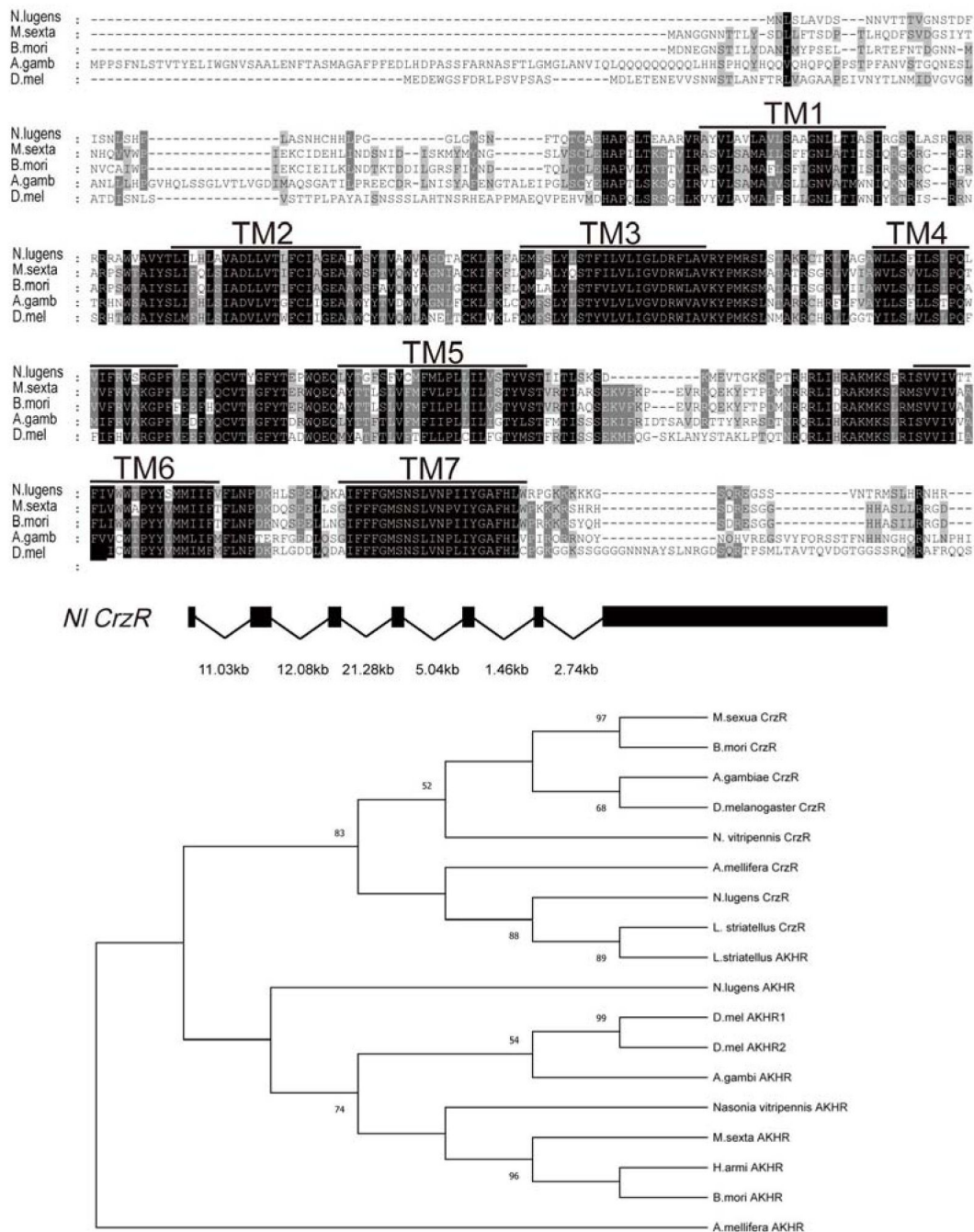


Figure 3 figure supplement 1.

Structural conservation of the *CrzR* across insect species.

(A) Multiple sequence alignment highlighting conserved transmembrane domains (TM1-TM7, shaded) in CRZ receptor orthologs. The species shown are: *N. lugens*: *Nilaparvata lugens*; *M. sexta*: *Manduca sexta*; *B. mori*: *Bombyx mori*; *A. gamb*: *Anopheles gambiae*; *D. mel*: *Drosophila melanogaster*. (B) Phylogenetic analysis of *CrzR* and *AkhR* based on the alignment of amino acid sequences of insect species. Phylogenetic tree was constructed using MEGA 12 software with the Maximum Likelihood Method and bootstrapped with 10000 replications. The numbers at the nodes of the branches represent the numbers of bootstrap replications supporting that branch.

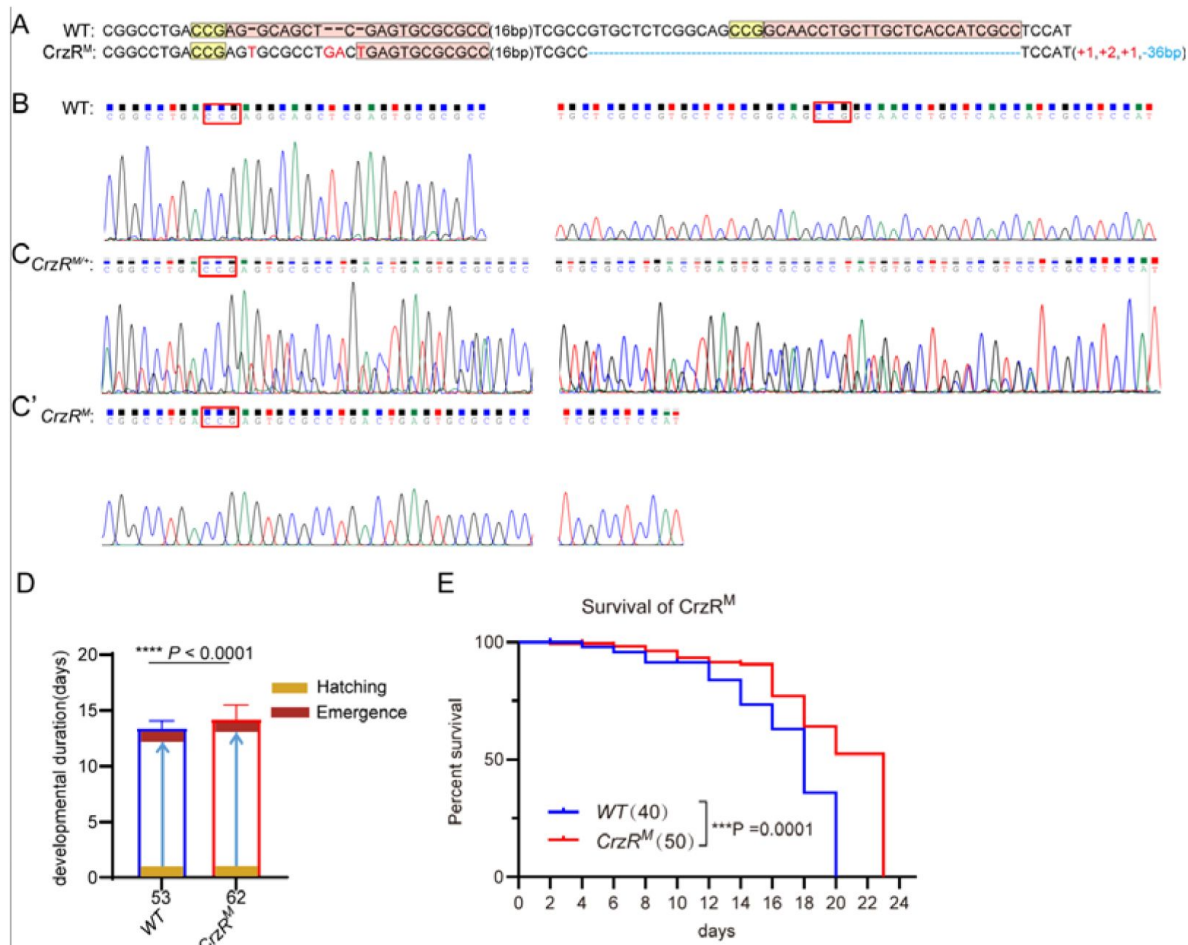


Figure 3 figure supplement 2.

Characterization of *CrzR* mutants in *N. lugens*.

(A) Sanger sequencing of *CrzR* mutants. WT sequence with target (pink), PAM (yellow). Indels: + (insertion), - (deletion). (B) Sanger sequencing chromatograms of *CrzR* alleles: WT (wild-type). (C) Sanger sequencing chromatograms of *CrzR* alleles: heterozygous (*CrzR*^{M/+}). (C') Sanger sequencing chromatograms of *CrzR* alleles: homozygous (*CrzR*^M) mutants. (D) Developmental duration (egg to adult eclosion) in *CrzR* mutants compared to WT. Data: mean ± SD. ****P < 0.0001 (Student's *t*-test; n ≥ 40/group). (E) Survival curves of adult *CrzR* mutants compared to WT. ***P < 0.001 (Log-rank Mantel-Cox test; df = 1; n ≥ 50/group).

The *CrzR* is highly expressed in the reproductive organs of female BPHs

To further elucidate the role of the CRZ signaling in regulation of the PMR, we analyzed the *CrzR* expression in *N. lugens* tissues. We first monitored *CrzR* expression by means of RT-PCR and qPCR. Both methods revealed predominant *CrzR* transcript expression in the female reproductive tract (Figure 4A-C). qPCR further demonstrated significantly higher *CrzR* expression in adult females than in males (Figure 4 - figure supplement 1A), particularly localized to female abdomens (Figure 4 - figure supplement 1B).

As we failed to generate a functional NlCrzR antibody, to determine the tissue expression of the receptor in more detail, we used *in situ* hybridization and confirmed strong *CrzR* expression within the female reproductive system (Figure 4D). No hybridization signal was detected in the lateral or common oviducts (Figure 4D1). However, specific labeling occurred in the sperm storage organs: the spermathecae and pouched glands of the lower reproductive tract (Figure 4D2 and 4D2'). These structures are the primary sites for long-term sperm storage. Specificity was validated by absence of signal in sense-probe controls (Figure 4D3 - D4'). This enrichment in sperm storage organs suggests that CRZ signaling may modulate post-mating physiology in part by regulating the mobilization of stored sperm and seminal fluid.

Notably, while *CrzR* is highly expressed in the female reproductive tract, we acknowledge that RT-PCR and qPCR analyses also detected low but detectable *CrzR* expression in other tissues, including the CNS and fat body (Figure 4B-C). Thus, we cannot exclude the possibility that *CrzR* in the brain (e.g., by modulating neural circuits governing reproductive behavior) or fat body (e.g., by affecting post-mating metabolism and nutrient/energy allocation) may also contribute to the PMR. These non-reproductive tract sites of *CrzR* expression warrant further investigation to fully delineate the systemic roles of CRZ/*CrzR* signaling in female PMR regulation.

CRZ signaling in male BPHs impacts the male accessory glands and secondarily modulates the female PMR

Although our primary focus is on CRZ signaling in the female PMR, we also investigated its role in male BPHs, given its established importance in male reproductive biology in *Drosophila* ^{44,46,48}. In *Drosophila*, silencing CRZ-expressing neurons in the male abdominal ganglia disrupts sperm/seminal fluid transfer and prolongs copulation, while their activation, or CRZ peptide injection, induces rapid ejaculation ⁴⁴. To test whether this function is conserved in BPHs, we first injected synthetic CRZ peptide into male BPHs. In contrast to *Drosophila*, CRZ injection did not trigger precocious ejaculation in male BPHs (Figure 5-figure supplement 1A) nor alter the morphology of the male accessory gland (MAG) (Figure 5-figure supplement 1B). However, CRZ-injected males exhibited a significantly reduced mating success rate (defined as the proportion of all tested individuals that successfully court and complete copulation), compared to controls injected with scrambled CRZ (sCRZ) (Figure 5A and B), but maintained unaltered courtship rates (courtship drive) and copulation durations (Figure 5-figure supplement 2A and B).

Since the male mating success rate in some cases was affected by CRZ, we made sure that the female PMR defects do not stem from incomplete male mating: all females included in our PMR analyses were mated with males that had completed full, successful copulation (i.e., completed all mating steps, including seminal fluid transfer, as verified by behavioral observation and subsequent validation of sperm presence in female reproductive tracts). Strikingly, even with fully successful mating, females mated with CRZ-injected males displayed a severely impaired PMR: nearly 30% of these females re-mated within the test window (vs. 0% of females mated with sCRZ-injected controls) (Figure 5B), and their oviposition was significantly reduced (Figure 5C).

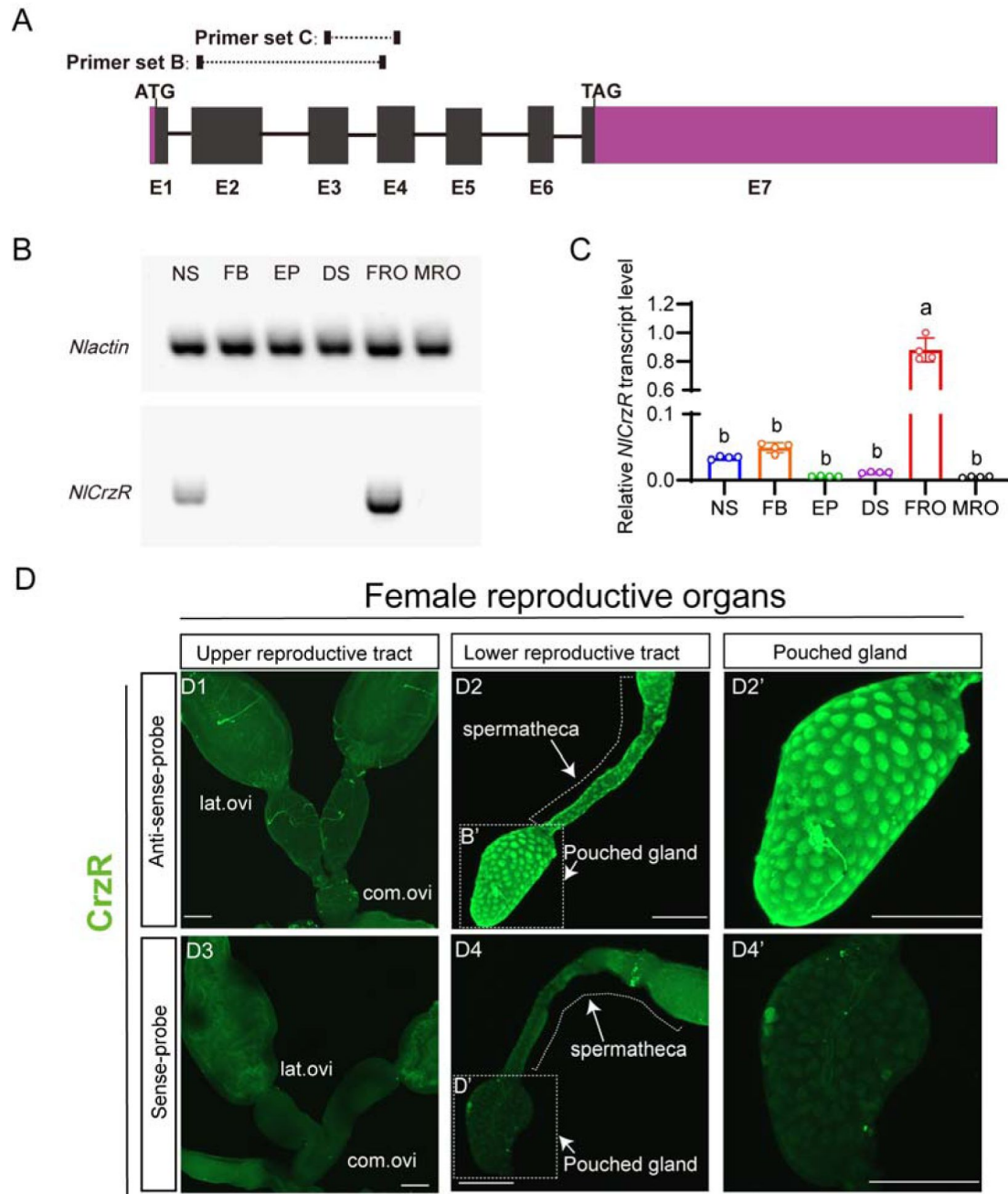


Figure 4.

Spatial expression profiling of *CrzR* in the female *N. lugens* reproductive tract.

(A) *CrzR* genomic structure with primer design for experiments in panels B-C. (B) RT-PCR analysis of *CrzR* tissue distribution. Note weaker expression in central nervous system and fat body. Actin loading control is shown in top row. Tissue abbreviations: NS: central nervous system; FB: fat body; EP: Epidermis; DS: digestive system, FRO: female reproductive organs, MRO: male reproductive organs. Except for the reproductive system, the rest of the tissues come from a mixture of female and male. (C) qPCR quantification of *CrzR* transcript levels (abbreviations as in B). Data are shown as mean $\pm$ s.e.m. Groups that share at least one letter are statistically indistinguishable. One-way ANOVA followed by Tukey's multiple comparisons test. (D1 - D2') *CrzR* mRNA localization via fluorescent *in situ* hybridization (antisense probe): D1: There is no signal in lateral/common oviducts; D2/D2': Specific signal is detected in spermatheca (SP) and pouched gland (PG). (D3 - D4') Negative controls (sense probe) showing background-level signal. Scale bars: 100 μ m.

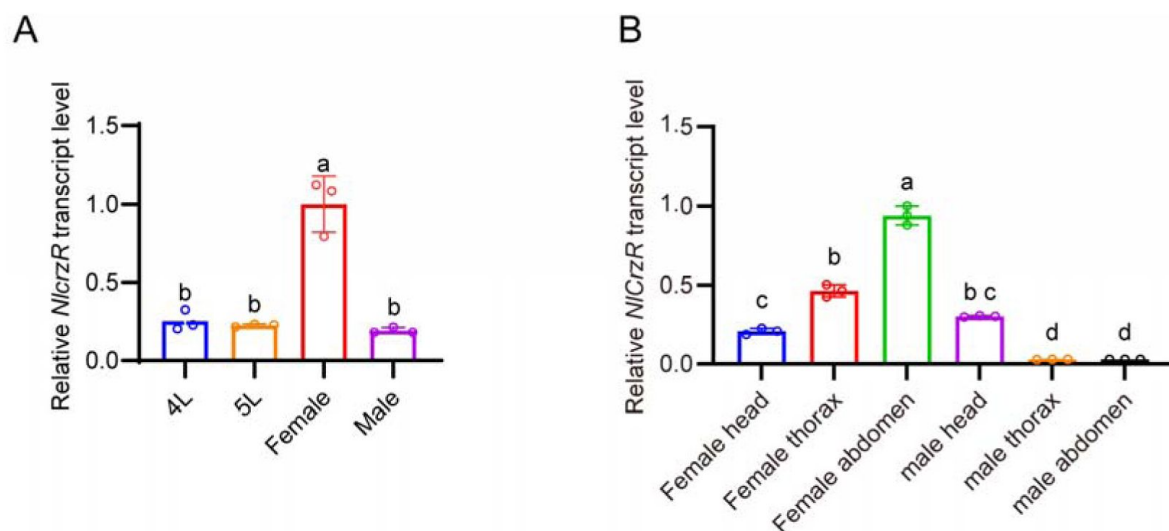


Figure 4 figure supplement 1.

Developmental and tissue-specific expression profiling of *CrzR* in *N. lugens* determined by qPCR.

(A) *CrzR* transcript levels across developmental stages. One-way ANOVA followed by Tukey's multiple comparisons test. (B) *CrzR* expression in distinct body regions. Data are shown as mean ± SEM shown. One-way ANOVA followed by Tukey's multiple comparisons test.

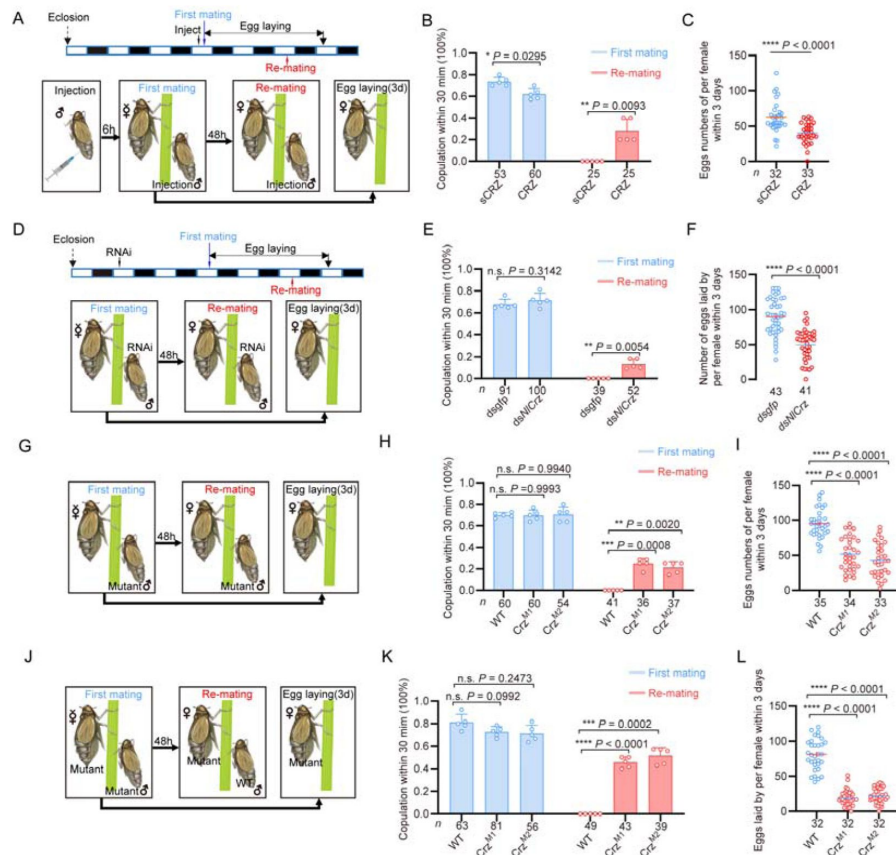


Figure 5.

CRZ signaling in male brown planthoppers affects the induction of PMR in females.

(A, D, G, J) Experimental protocols for the behavioral assays presented in panels B/C, E/F, H/I, and K/L, respectively. (B) Receptivity of virgin and mated wild-type females, scored as the percentage copulating within 30 min when paired with males injected with CRZ (sCRZ as control). Only the re-mating is affected. Small circles indicate the number of replicates; numbers below bars denote total animals. Data are mean $\pm$ s.e.m. * P < 0.05, ** P < 0.01 (two-way ANOVA followed by Holm-Šidák's multiple comparisons test). (C) Number of eggs laid per wild-type female within 72 h after mating with males injected with CRZ (sCRZ as control). Numbers below bars denote total animals. Data are mean $\pm$ s.e.m. *** P < 0.001 (Student's t -test). Experiments involved ≥ 4 biological replicates with ≥ 8 insects per replicate. (E) Receptivity of virgin and mated wild-type females, scored as the percentage copulating within 30 min when paired with *Crz*-RNAi-treated males (dsGFP as control). Again, only re-mating is affected. Small circles indicate the number of replicates; numbers below bars denote total animals. Data are mean $\pm$ s.e.m. ** P < 0.01, ns (P > 0.05) (two-way ANOVA followed by Holm-Šidák's multiple comparisons test). (F) Number of eggs laid per wild-type female within 72 h after mating with *Crz*-RNAi-treated males (dsGFP as control). Data are mean $\pm$ s.e.m. **** P < 0.0001 (Student's t -test). Experiments involved ≥ 4 biological replicates with ≥ 8 insects per replicate. (H) Receptivity of virgin and mated wild-type females, scored as the percentage copulating within 30 min when paired with males of wild type (WT) and CRZ mutants (ΔCrz^1 , ΔCrz^2 and $\Delta Crz^{1/2}$). Small circles indicate the number of replicates; numbers below bars denote total animals. Data are mean $\pm$ SEM. ** P < 0.01, *** P < 0.001, ns (P > 0.05) (two-way ANOVA followed by Holm-Šidák's multiple comparisons test). (I) Number of eggs laid per wild-type female within 72 h after mating with males of different genotypes (as in H). Numbers below bars denote total animals. Data are mean $\pm$ SEM. **** P < 0.0001 (Student's t -test). Experiments involved ≥ 4 biological replicates with ≥ 8 insects per replicate. (K) Crossing male and female *Crz* mutant insects yield a stronger phenotype than when only males are mutants. Receptivity of virgin and mated females when using *Crz* mutants of both males and females, with wild-type as controls, scored as the percentage copulating within 30 min when paired with males of different genotypes. Compared to Fig. 2H and 2I, double *Crz* knockouts exhibit a stronger effect on receptivity. Small circles indicate the number of replicates; numbers below bars denote total animals. Data are mean $\pm$ SEM. *** P < 0.001, **** P < 0.0001, ns (P > 0.05) (two-way ANOVA followed by Holm-Šidák's multiple comparisons test). (L) Number of eggs laid per female in different genotypic mating combinations. Again phenotypes are stronger compared to those observed with only male mutants (Fig. 2I). Numbers below bars denote total animals. Data are mean $\pm$ SEM. **** P < 0.0001 (Student's t -test). Experiments involved ≥ 4 biological replicates with ≥ 8 insects per replicate.

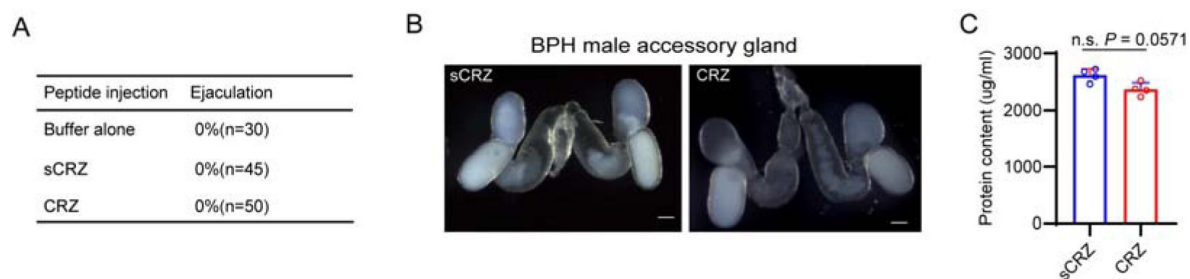


Figure 5 figure supplement 1

The effects of CRZ neuropeptide injection on the behavior and physiology of male BPHs.

(A) CRZ peptide injection into the terminal abdomen of male BPHs does not affect ejaculation. The table shows the ejaculation frequency within 30 minutes post-injection. (B) Morphology of male BPH accessory glands following *Crz* gene knockout. Representative images show accessory glands from *Crz* mutants and wild-type (WT) controls. (C) Accessory gland protein content (determined by BCA method assay) after CRZ neuropeptide injection (sCRZ control). Data show mean $\pm$ SEM; ns ($P > 0.05$, Mann-Whitney *U* test).

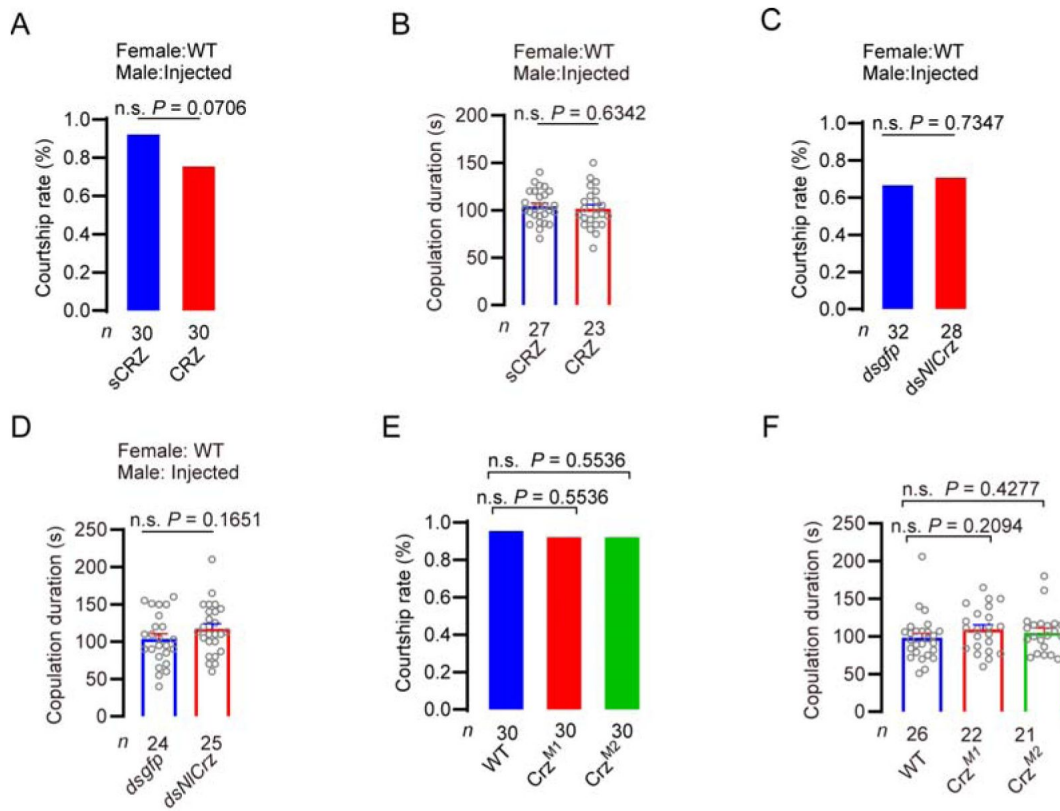


Figure 5 figure supplement 2.

Manipulation of *Crz* in male BPHs does not affect male courtship behavior.

(A) Courtship rate (number of males that exhibit courtship behavior) in males after CRZ peptide injection (sCRZ control) and mating with wild type (WT) females. Numbers below bars denote total animals. Data represent mean; ns ($P > 0.05$, χ^2 test). (B) Mating duration of CRZ-injected males paired with WT females (sCRZ control). Data show mean $\pm$ SEM; ns ($P > 0.05$, Student's *t*-test). (C) Courtship rate after *Crz* knockdown via RNAi in males (dsGFP control). Data represent mean; ns ($P > 0.05$, χ^2 test). (D) Mating duration of *Crz*-knockdown males paired with wild-type females (dsGFP control). Data show mean $\pm$ SEM; ns ($P > 0.05$, Student's *t*-test). (E) Courtship rate of *Crz* mutant males (WT controls). Data represent mean; ns ($P > 0.05$, χ^2 test). (F) Mating duration of males of different *Crz* mutants paired with WT females. Data show mean $\pm$ SEM; ns ($P > 0.05$, Student's *t*-test).

Although CRZ injection did not quantitatively change the total protein content of the MAG (**Figure 5-figure supplement 1C**), we hypothesize that it alters the composition of seminal fluid proteins, consistent with findings in the oriental fruit moth *Grapholita molesta*⁴⁹ and thereby reducing the ability to induce a robust female PMR. In male BPHs, RNAi-mediated knockdown of the *Crz* transcript (via dsNlCrz injection) had no significant effect on key male reproductive behavior: mating success rate (proportion of males that complete copulation), courtship rate (proportion of males exhibiting courtship behaviors such as following or wing display), and copulation duration (**Figure 5D** and **E**, **Figure 5—figure supplement 2C** and **D**). However, females mated with dsNlCrz-injected males exhibited a defective PMR: about 10% remated (**Figure 5E**) and laid approximately 60% fewer eggs (**Figure 5F**). Consistent with the RNAi results, males carrying *Crz* mutant alleles (ΔCrz^1 or ΔCrz^2) also failed to elicit a full PMR in mated females, despite displaying normal courtship behavior and mating success (**Figure 5—figure supplement 2E** and **F**). Specifically, ~25% of the females mated with *Crz* mutant males accepted re-mating (**Figure 5G and H**), and their egg production was halved compared to females mated with wild-type males (**Figure 5I**).

Notably, when *Crz* mutant males were crossed with *Crz* mutant females, the PMR defects were dramatically exacerbated: the re-mating rate of females surged to nearly 50%, and egg production plummeted by ~75% relative to wild-type male-female pairs (**Figure 5J** - **L**). This near-complete ablation of the PMR underscores that intact CRZ signaling in both sexes is indispensable for triggering a robust PMR in female BPHs.

CRZ signaling in male BPHs alters seminal fluid composition

The observation that manipulating *Crz* (via knockdown or mutation) in male BPHs affects the female PMR is particularly intriguing, prompting us to investigate the underlying mechanism. Since we had ruled out production of CRZ in the male reproductive system, including the male accessory gland (MAG), we set out to test whether males with impaired CRZ signaling exhibit defects in basic reproductive performance (e.g., mating success, sperm production/transfer).

To eliminate confounding effects of abnormal sperm production or transfer, we first assessed some key parameters: spermathecal size in females mated to wild-type (WT) or ΔCrz^1 males showed no significant differences (**Figure 6A**), and sperm counts in the testes of unmated ΔCrz^1 males (vs. WT) and in the spermathecae of females mated to ΔCrz^1 males (vs. WT) were comparable (**Figure 6B**). These data confirm intact sperm production and transfer, pointing to potential alterations in seminal fluid composition as the driver of female PMR defects. This led us to focus on the protein production in the MAG, encouraged by prior work in the oriental fruit moth (*G. molesta*) where it was shown that CRZ signaling modulates biosynthetic activity in the MAG⁴⁹.

To approach this, we performed RNA-sequencing (RNA-seq) on MAGs from WT and ΔCrz^1 males. Comparative transcriptomics identified 2,547 differentially expressed genes (DEGs) in ΔCrz^1 MAGs ($|\log_2\text{FC}| > 1$, $\text{FDR} < 0.05$), with 2,114 upregulated and 431 downregulated (**Figure 6C**). Notably, 16 genes encoding seminal fluid proteins (SFPs) were dysregulated (7 downregulated, 9 upregulated). Gene Ontology (GO) enrichment analysis of downregulated DEGs revealed significant overrepresentation of terms related to serine-type endopeptidase activity (GO: 0004252; $\text{Padj} < 0.05$)—a function critical for SFP processing (**Figure 6D**).

Consistent with transcriptional dysregulation, the total protein content of ΔCrz^1 male MAGs was significantly lower than that of WT MAGs, as quantified by the Bicinchoninic Acid (BCA) assay (**Figure 6E**). Targeted qPCR of specific genes in the MAG confirmed significant downregulation of two genes (**Figure 6F and G**), and also confirmed unchanged transcript levels of the MAG peptides, *macc* and *ITPL-1*, in mutants (**Figure 6H and I**). In summary, the transcriptome

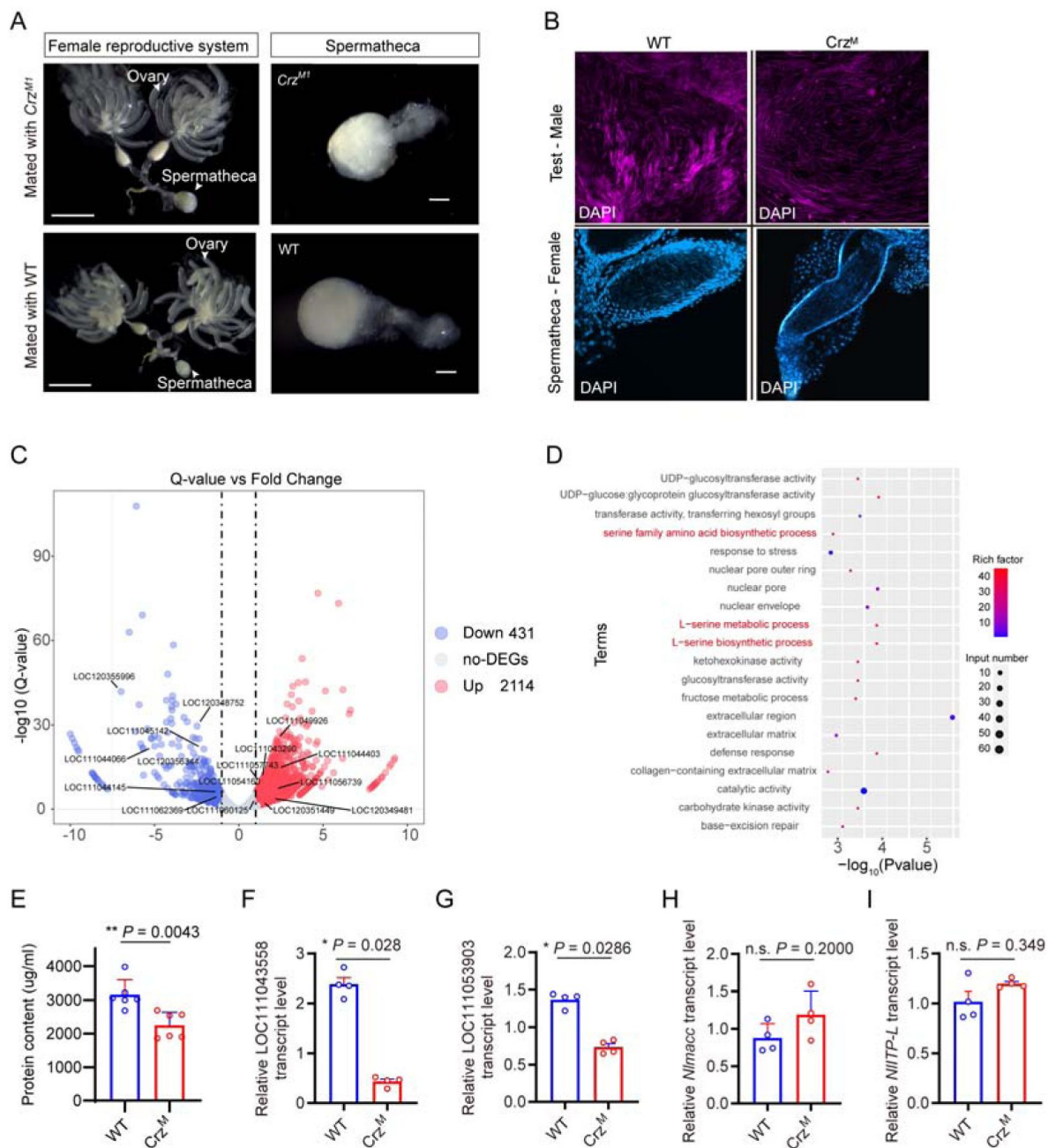


Figure 6.

CRZ signaling affects seminal fluid protein composition in male BPHs

(A) Spermathecae of WT females mated to WT or ΔCrz^1 males. Scale bar: 1 mm. No morphological differences were observed ($n = 15$ pairs) (B) DAPI-stained testes (males) and spermathecae (females mated within 4 h) of WT or ΔCrz^1 BPHs. Scale bar: 100 μm . Sperm counts showed no genotype-dependent differences (testes: $P = 0.32$; spermathecae: $P = 0.41$; $n = 12$; Mann-Whitney U test). (C) Volcano plot of differentially expressed genes in male accessory glands (MAGs) from ΔCrz^1 mutants versus wild-type (WT) males. Seminal protein genes with significant differential expression ($|\log_2\text{FC}| > 1$, $P < 0.05$) are highlighted (black line). Gene IDs for seminal proteins are annotated. Upregulated genes are shown in red and downregulated in black. (D) Top 20 enriched Gene Ontology (GO) terms among downregulated genes. Rich factor indicates enrichment strength; input number denotes enriched genes per term. (E) MAG protein content in WT and ΔCrz^1 males measured by BCA method assay. Data: mean $\pm$ SEM; $**P < 0.01$ (Mann-Whitney U test; $n = 8$ MAG pairs). (F-I) qPCR validation of seminal fluid genes 43558, 53903, macc and ITP-L in MAGs. Data: mean $\pm$ SEM ($n = 4$ biological replicates). ns: not significant ($P > 0.05$); $*P < 0.05$ (Mann-Whitney U test).

analysis indicates that CRZ influences the expression of MAG genes and this results in a defective PMR in mated females. Since *macc* and *ITPL-1* are not among the CRZ-regulated genes, some other gene(s) encoding PMR inducing factors need to be identified in future work.

Endogenous CRZ signaling modulates aspects of the PMR in female

Drosophila melanogaster

Having established the critical role of CRZ signaling in regulating the PMR of the female BPH, we next investigated its potential function in female reproduction, including the PMR, of the genetically tractable fly *Drosophila melanogaster*.

We first assessed re-mating frequency in *Crz* mutant female flies (*Crz^{attp}*). Approximately 40% of the *Crz^{attp}* females that had experienced a first successful copulation mated again, compared to only 3% of mated wild-type control females (**Figure 7A and B**). Importantly, the *Crz* mutation did not affect the first mating of females (**Figure 7B**). Furthermore, mated *Crz^{attp}* females laid approximately 45% fewer eggs than mated wild-type females (**Figure 7C**). These results demonstrate that CRZ signaling is necessary for aspects of the PMR in female *Drosophila*, specifically inhibiting re-mating and stimulating post-mating egg production. Immunohistochemical labeling confirmed the absence of CRZ protein in *Crz^{attp}* mutants (**Figure 7D**).

To determine whether release of CRZ from *Crz* neurons is responsible for inhibition of remating and stimulation of oviposition, we knocked down *Crz* expression specifically in CRZ-producing neurons using the GAL4-UAS system (*Crz-GAL4 > UAS-Crz-RNAi*). Consistent with the mutant phenotype, 22% of the mated *Crz-GAL4 > UAS-Crz-RNAi* females mated again after their first successful mating with wild-type males, whereas only 2% of the control females (*Crz-GAL4/+*) did so (**Figure 7E**). Virgin receptivity was again unaffected by *Crz* knockdown (**Figure 7E**). Additionally, mated *Crz-GAL4 > UAS-Crz-RNAi* females laid approximately 20% fewer eggs than mated controls (*Crz-GAL4/+*) (**Figure 7F**). To further confirm the role of CRZ neuropeptide in modulation of the PMR in females, we injected this peptide in virgin females. Consistent with the above results, CRZ injected virgins displayed a reduced response to mating attempts (**Figure 7G**). In conclusion, like in BPHs, the neuropeptide CRZ endogenously modulates post-mating behavior also in female *Drosophila*.

Discussion

Our study unveils a novel function of CRZ in an endogenous female signaling pathway that modulates the post-mating reproductive behavior and physiology (collectively PMR) in BPHs. It is, thus, the first report linking CRZ to a PMR in insects. In contrast to SP and DUP99B in *Drosophila* ^{5,18}, HP-1 in *Aedes aegypti* ¹⁵ and maccessin (*macc*) recently identified in BPHs ²⁵, CRZ is not produced in the MAG, and thus not transferred from males during copulation. Instead, it is acting as an endogenous peptide derived from neuroendocrine cells in the female brain. Moreover, this CRZ-mediated endogenous modulation in females is seen also in *Drosophila*. When *Crz* is knocked down or knocked out in flies, mated females likewise exhibit an altered PMR.

Interestingly, we found that in BPH males, endogenous CRZ acts on the MAG to affect biosynthesis of seminal fluid proteins (SFP), similar to another insect, the lepidopteran *Grapholita molesta* ⁴⁹. That study revealed that silencing the *Crz* gene results in a downregulation of multiple genes in the MAG, including several serine-type endopeptidases, however, the authors did not analyze the effect on the female PMR. Our experiments demonstrate that with impaired CRZ signaling in males the seminal fluid appears to lack some component(s) necessary for inducing a PMR since females

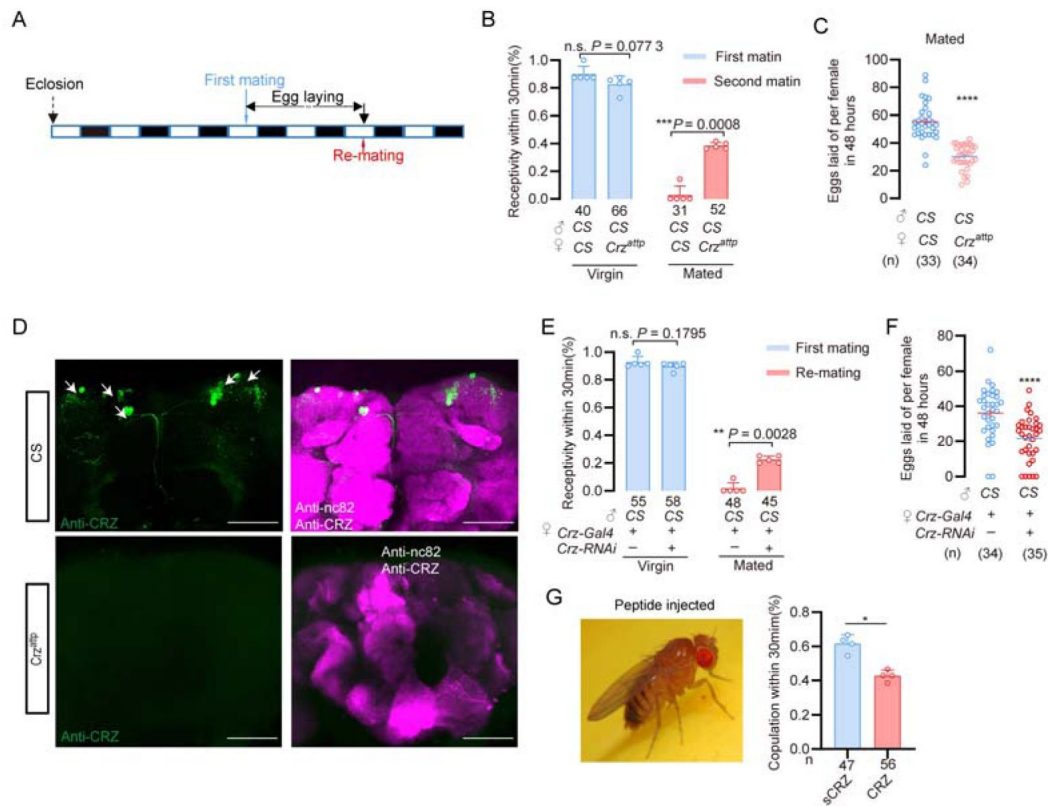


Figure 7.

CRZ regulates part of the PMR in *Drosophila melanogaster*.

(A) Schematic of experimental design. White segments on the time axis indicate light periods; black segments indicate dark periods. (B) Receptivity of virgin and mated *Crz* mutant (*Crz^{attp}*) females, scored as the percentage copulating within 30 min. *** $P < 0.001$ (Chi-square test). (C) Mated *Crz* mutants (*Crz^{attp}*) lay more eggs. Graph shows eggs laid per female after mating. **** $P < 0.0001$ (Student's *t*-test). (D) Number of eggs laid per virgin female *Crz* mutants, n.s.: $P > 0.05$ (Student's *t*-test). (E) Anti-CRZ immunostaining in brains of *Crz* mutant (*Crz^{attp}*), control (CS) and after *Crz*-RNAi. Note CRZ-expressing neurons at arrows in control. Scale bar: 100 μ m. (F) Receptivity of virgin and mated females after *Crz*-RNAi. **** $P < 0.0001$; Chi square test. (G) *Crz*-RNAi reduces the number of eggs laid by mated females. **** $P < 0.0001$ (Student's *t*-test). (H) Receptivity of virgin females following CRZ peptide injection (monitored 6 h after injection). * $P < 0.05$ (Mann-Whitney test).

mated by *Crz* knockdown males display a reduced PMR. Our gene expression data reveals that neither *macc* nor *ITPL-1* are among the MAG genes affected by *Crz* knockout. These are encoding peptides that are known to induce a partial PMR in female BPHs ²⁵. Our results therefore suggest that there are additional PMR-inducing factors produced by the MAG of BPHs and that these are regulated by CRZ. Further work is required to identify these factors, but from the present data we can conclude that in BPHs, CRZ signaling influences the female PMR both by action in males and females, but in distinct ways. Similar to our CRZ findings here, a possible dual role of ITPL-1 was also seen in BPHs ²⁵. This peptide is produced in the MAG, but also in tissues of female BPHs, and knockdown in males results in a reduced PMR in females ²⁵.

Focusing on the endogenous CRZ signaling in female BPHs, our main findings using mutants and RNAi of both *Crz* and *CrzR*, demonstrate that CRZ signaling decreases the female responsiveness to courting males, especially in already mated females. Also, oviposition in mated BPHs is increased by CRZ signaling. CRZ immunolabeling revealed a small set of neurons in the brain and subesophageal ganglion and no CRZ was detected in reproductive organs. Thus, the likely source of CRZ affecting the PMR is the CRZ-producing neuroendocrine cells in the brain. In insects studied so far (including other hemipterans), there are at least three pairs of CRZ expressing lateral neurosecretory cells with axon terminations in the corpora cardiaca ^{52,53}, suggesting that in BPHs and *Drosophila* hormonal CRZ may contribute to the modulation of the PMR. However local CRZ signaling in brain circuits cannot be excluded since the CRZ neurons arborize extensively within the brain.

We found that the CRZ receptor, *CrzR*, is predominantly expressed in the female reproductive tract in BPHs, more specifically in the lower tract including the spermatheca and pouched gland, which are sites of sperm and seminal fluid storage. Thus, in BPHs, CRZ may act to regulate the release of stored sperm and seminal fluid in the female. This regulation would also gate the flow of male-derived factors that induce and maintain the PMR, such as *macc* and ITPL-1. As mentioned, we cannot exclude additional action of CRZ in neuronal circuits in the CNS that regulate female sexual arousal and reproductive behavior, since our PCR data indicate expression of the *CrzR* also in the CNS. Additionally, the fat body of BPHs, express *CrzR*, similar to *Drosophila* ⁵⁴, possibly suggesting a role of CRZ in post-mating metabolism.

The brain CRZ neurons of BPHs are likely to be activated by sensory signals from the female reproductive tract upon copulation and hence release CRZ into the circulation (and maybe within the brain). In *Drosophila*, sensory neurons in the reproductive tract are activated by SP (via the SP receptor), and signal via a chain of ascending axons of the VNC to sex-specific neurons in the brain to induce the PMR switch in behavior and physiology ^{7,35,37,55}. Thus, in *Drosophila*, these sex-specific circuits may also activate the CRZ neurons. Possibly similar sensory and sex-specific neurons exist in BPHs and respond either to mechanical stimuli or factors in the seminal fluid leading to activation of CRZ neurons and other brain neurons that regulate the PMR.

The female PMR is complex in *Drosophila* and not only receptivity and oviposition are affected, but also feeding, metabolism, sleep patterns and aggression ^{5,16,18,19,56,57}. In BPHs only the receptivity (and associated rejection behavior) and oviposition have been analyzed in any detail. We found that the MAG-derived peptides transferred with the semen, *macc* and ITPL-1, only affected the receptivity ²⁵, whereas we show here that CRZ affects also oviposition. Thus, we have identified three different peptides CRZ, *macc* and ITPL-1 that all influence the PMR in different ways, suggesting complex regulatory mechanisms also in BPHs.

It is interesting to note that CRZ is an evolutionarily conserved neuropeptide that has been found in most, but not all, insect species and it displays a wide array of functions ^{53,54,58,59}. The CRZ function varies across different insect species, but a common role seems to be in regulation of stress ⁵⁹⁻⁶¹. Other functions are for example: in the American cockroach, CRZ accelerates the heart rate ⁶², in locusts, it triggers the formation of pigments that darken the body color in

gregarious insects⁶³, in the moth *Bombyx mori*, CRZ affects developmental speed and reduces the rate of silk spinning behavior⁶⁴; in *Drosophila* larvae, CRZ regulates molting motor behavior⁶⁵ and growth⁶⁶; in adult *Drosophila* CRZ regulates stress responses, food intake and metabolism^{54,67}, and protein food preference¹², as well as sperm ejection and mating duration in males⁴²; and in social ants, it regulates the social hierarchy and inhibits egg-laying in the queen⁴⁵. Finally, a recent study found that CRZ mediates a diapause-induced suppression of reproduction in female bean bugs⁶⁸. To all these functions can now be added the role of CRZ in reproductive behavior and physiology of BPHs, especially in the female PMR of BPHs and *Drosophila*. However, we also found a role of CRZ signaling in regulating the production of seminal fluid proteins in the MAG. Finally, building on the role of CRZ in metabolism in *Drosophila*⁵⁴, it is possible that CRZ also regulates post-mating metabolism in female BPHs. Thus, there may be a link between CRZ and the altered feeding and metabolism commonly seen in mated female insects.

Key resources table

REAGENT or RESOURCE	Source or reference	Identifiers
Antibodies		
Goat anti-Rabbit Alexa 488	Invitrogen	Cat#A11008; RRID: AB_143165
Goat anti-mouse Alexa 488	Invitrogen	Cat#A28175; RRID: AB_2536161
Goat anti-mouse Alexa 555	Invitrogen	Cat# A21428; RRID: AB_2535849
Anti-CRZ	Gift from Prof. Jan A. Veenstra	
Chemicals, Peptides, and Recombinant Proteins		
4% PFA	Leagene	Cat# DF0134
DAPI	Solarbio	Cat# C0060
PBS	Solarbio	Cat# P1022
ReadyProbes 2.5% Normal Goat Serum	Invitrogen	Cat# R37624
Experimental Models: Organisms/Strains		
<i>Canton-S</i>	Bloomington Drosophila Stock Center	64349
<i>Crz^{attP}</i>	Rao and Deng et al., 2019	BDSC: 84487
<i>Crz-GAL4</i>	Bloomington Drosophila Stock Center	51976
<i>Crz-RNAi</i>	Tsinghua Fly Center	THU2139
Software and Algorithms		
FIJI		https://doi.org/10.1038/nmeth.2019
Prism 9	GraphPad	RRID: SCR_002798
Photoshop	Adobe	RRID: SCR_014199
Illustrator	Adobe	RRID: SCR_010279
ZEN	Zeiss	N/A
Motic Images Plus3.0	Motic	N/A
Motic Images Devices	Motic	N/A
Other		
Egg-laying device	Gou et al., 2016	N/A
Anatomical tweezers	Dumont	Cat# 0209-5-PO

Method details

Experiment insects

All the brown planthoppers used in this research were collected from Hangzhou, Zhejiang Province in 1995. The insects were kept indoors on fresh rice seedlings (Taichung Native 1, TN1) in a growth chamber at $27 \pm 1^\circ\text{C}$ and $70 \pm 10\%$ relative humidity, under a 16 h:8 h (Light: Dark) photoperiod ⁶⁹[\[69\]](#).

All *Drosophila* stocks were raised in standard cornmeal–molasses–agar medium maintained at 25°C , 60% humidity, and under 12 hr:12 hr light:dark conditions. We used Canton s flies as the wild-type strain. The relevant information of fruit flies used in the experiments are shown in key resources table.

Behavior assays of *Nilaparvata lugens*

Mating and receptivity

All BPHs were raised in incubators with a 16 h:8 h dark:light cycle. Unmated female and male BPHs were collected within 12 hours of emergence. For RNAi or neuropeptide injection treatment, the protocol is shown in **Figure 1A** [\[69\]](#) and **Figure 1D** [\[69\]](#). The mating behavior (or female receptivity behavior) test in this study usually conducted the first mating between 3 and 5 days after emergence. The mating experiment was completed in a plastic tube (2.5cm in diameter and 10cm in height). Fresh rice seedlings of appropriate length were placed in each tube, and then a female and male BPHs were added. The mating conditions and other reproductive behavior parameters during this period were observed in a constant temperature room at 25°C for 30min, and the males were taken out after 30 min. For the experiment requiring a second mating, females successfully mated for the first time were retained. After 48 hours, wild-type males were put in for the re-mating test. The males in all secondary mating experiments were wild type. The monitoring time for each mating behavior was 30 minutes; the receptivity rate refers to the proportion of successful matings of female individuals within the 30-minute measurement period.

Egg laying

After completing the first mating, the male is sucked out and a sufficient number of rice seedlings are added to the tube, and the female is retained to lay eggs in the tube. After 72 hours, the spawning female is transferred out of the tube. After the eggs have matured, about 4-5 days later, we used tweezers to gently manipulate the rice seedlings and counted the number of fertilized eggs produced by the females. Note that 4 to 5 days after the egg is produced, the successfully fertilized egg will display red eye-spots. (Oviposition assay protocol: For quantitative assessment of post-mating responses, egg-laying activity was monitored over a 72-hour observation window following verified initial copulation. To eliminate confounding effects from sequential mating trials, experimental subjects undergoing oviposition measurement were permanently excluded from subsequent re-mating evaluations.)

To determine the oviposition profile of females injected with CRZ peptide, virgin female or mated female BPHs that had emerged 3-5 days earlier were transferred into plastic tubes with sufficient amount of rice seedlings 6h after recovery from CRZ injection, and the BPHs were taken out 24 hours later, using tweezers to gently dissect the rice seedlings the number of fertilized eggs produced by the females were counted.

Courtship rate

During the mating assessment, an unmated (virgin) female and an unmated male BPH was placed in a plastic tube containing rice seedlings. The occurrence of courtship behavior in the male individuals was recorded, which included a series of courtship actions such as following, wing

display, licking, and attempting to mate. The proportion of individuals displaying courtship behavior among all experimental individuals is defined as the courtship rate.

Copulation duration

During the mating assessment, an unmated (virgin) female and an unmated male BPH were placed in a plastic tube containing rice seedlings. Timing begins when mating behavior is observed between the female and male, and stops immediately after completed copulation. The duration from the start of the mating behavior to its conclusion is defined as the mating duration.

Behavior assays of *Drosophila melanogaster*

Mating and re-mating : During the receptivity assessment, all experimental fruit flies were virgin flies that emerged within 12 hours in a 20°C incubator in a 12 h:12 h dark:light cycle. After collection, they were kept at 25°C incubator in a 12 h:12 h dark:light cycle for 4 days before behavioral testing. **Receptivity behavior measurement :** a perforated plate was placed on top of a non-perforated plate. Male fruit flies were briefly anesthetized on ice and gently transferred into the holes of the perforated plate, with one male per hole. Once the male transfer was completed, a plastic film was gently placed over the perforated plate, followed by covering it with another perforated plate. Then, the virgin female fruit fly, also anesthetized on ice, was placed in each hole. After the transfer was completed, a non-perforated plate was placed on top. This setup was then transferred to a 25°C incubator to acclimate for 30 minutes. The plastic film separating the male and female flies was quickly removed, and video recording began for 30 minutes. The recorded videos were analyzed to calculate the acceptance rate of the females. Successful mating females were collected and individually transferred into plastic tubes containing sufficient artificial food for fruit flies to lay eggs. After 24 hours of laying eggs, the female flies were transferred to a new finger-shaped tube with ample artificial feed for another 24 hours of laying. After 48 hours, these egg-laying females were subjected to a secondary mating assessment with wild-type, unmated males, using the same method as described above, to evaluate the acceptance rate during the second mating process. Additionally, the number of eggs laid by each female during the 48 hours was recorded.

Egg laying for mated females : First, collect virgin female fruit flies that eclosion within 12 hours at 20°C incubator in a 12 h:12 h dark:light cycle and then keep them in a 25°C incubator in a 12 h:12 h dark:light cycle for 4 days. After that, unmated males of the wild type and virgin females were mixed 1:1.5 times. 24 hours after full mating, female flies were lightly anesthetized with ice and transferred to plastic vials containing enough artificial fruit fly food for them to lay eggs for 24 hours. After 24 hours, the flies were transferred again to a new bottle full of artificial food for another 24 hours of egg laying. The number of eggs in each vial is calculated under a microscope, and the total number of eggs in the two vials of each female fruit fly represents the egg-laying capacity of that individual.

Developmental duration statistics

Nymphs of BPHs that hatched synchronously from distinct genotypes were collected and reared in plastic cups containing fresh rice seedlings; the seedlings were replaced regularly to maintain optimal conditions. From the day the nymphs reached the fifth instar, cups were inspected daily for adult emergence, and the number of newly emerged adults was recorded. After all surviving individuals had completed eclosion, the total nymph-to-adult developmental duration was calculated.

BPHs survival rate statistics

For neuropeptide injection: After 4-day emergence, virgin BPHs were injected with CRZ neuropeptide or scrambled CRZ, and after 6 hours of recovery, these insects were placed into a plastic tube with sufficient quantity of rice seedlings. Survival was monitored every 24 hours until

all specimens were dead.

For mutant: Wild-type and mutant BPHs adults that eclosed on the same day were individually confined in plastic tubes, each supplied with ample fresh rice seedlings and water renewed regularly to ensure optimal living and nutritional conditions. The number of surviving individuals of each genotype was recorded every 24 h until all had died, and survival curves were generated from these data.

Neuropeptide synthesis and injection

Corazonin (CRZ) and scrambled CRZ (sCRZ) were synthesized by Genscript Co. Ltd (Nanjing, China). Peptide masses were confirmed by Mass spectrometry and the amount of peptide was quantified by amino acid analysis. The amino acid sequences of the peptides used in this study are: *N. lugens* corazonin: pQTFQYSRGWNamide; scrambled corazonin: pQSRYTTFGQWTNamide. On the fourth day after adult emergence, virgin female BPHs were injected with different concentrations of synthetic CRZ peptide and control scrambled peptide. The scrambled peptide control was used to monitor for potential non-specific effects on behavior due to the process of injection. Each mg peptide was dissolved in 100 μ l DMSO solution (Acme, #D54264), and then diluted to 1.5 mM, 0.15 mM, 0.015 mM and 0.0015 mM with 1 $\times$ PBS as solvent. The BPHs were anesthetized with CO₂ for about 30 seconds and placed abdomen up on 2% agarose (Solarbio, #A8201) plate and 50 nl neuropeptide solution injected into each female, 30 nl neuropeptide solution injected into each male. The male virgins were injected on the third day of emergence. Six hours after neuropeptide injection the insects were used for behavior assay.

Protein concentration determination of male accessory glands

MAGs of 3-day-old unmated males (n = 48) were dissected in 1 $\times$ PBS (Solarbio, #P1020) and then gently crushed in 60 μ L of 1 $\times$ PBS. The tissues were homogenized for 60 s and centrifuged at 10,000g for 15 min at 4° C and the supernatant removed. The protein concentration in the supernatant was determined using the BCA Protein Assay Kit (CWBI, #CW0014S).

Gene cloning and sequence analysis

We used the NCBI database with BLAST programs to carry out sequence alignment and analysis. Then we predicted Open Reading Frames (ORFs) with EditSeq. The primers were designed by Primer designing tool in NCBI. Total RNA Extraction was using the TRIzol reagent (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions. The cDNA template used for cloning was synthesized using the Biotech M-MLV reverse transcription kit and the synthesized cDNA template was stored at -20°C. The transmembrane segments and topology of proteins were predicted by TMHMM v2.0 (<http://www.cbs.dtu.dk/services/TMHMM-2.0/>)⁷⁰. Multiple alignments of the complete amino acid sequences were performed with Clustal Omega (<http://www.ebi.ac.uk/Tools/msa/clustalo>).

qRT-PCR for spatial expression pattern and RNAi efficiency

The nervous system (NS), fat body (Fb), digestive system (DS), epidermis (EP), male reproductive organs (MRO) and female reproductive organs (FRO) from 3-5 days old adults were dissected in 1 $\times$ PBS, to determine the tissue-specific expression pattern. With the exception of the reproductive system, the rest of the tissue comes from both sexes equally. The adult BPHs were placed on ice under slight anesthesia and dissected under a stereoscopic microscope (ZEISS). At least 50 individuals were dissected as one sample for tissue-specific analysis with three replicates. All tissues were quickly transferred to Trizol (Invitrogen, Carlsbad, CA, USA) solution at 0° C after dissection.

For RNAi efficiency testing, we collected the whole bodies of BPHs and transferred to a 1.5 mL EP tube containing Trizol reagent. Total RNA extraction was using Trizol reagent (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions. The cDNA template used for cloning was synthesized using the Biotech M-MLV reverse transcription kit and the synthesized cDNA template was stored at -20°C.

Real-time qPCRs of the various samples used the UltraSYBR Mixture (with ROX) Kit (CWBIO, Beijing, China). The PCR was performed in 20 µl reaction including 2 µl of 10-fold diluted cDNA, 1 µl of each primer (10 µM), 10 µl 2 × UltraSYBR Mixture, and 6 µl RNase-free water. The PCR conditions used were as follows: initial incubation at 95°C for 10 min, followed by 40 cycles of 95°C for 10 s and 60°C for 45 s. *N. lugens* 18S rRNA and actin were used as an internal control (primers listed in Table1). Relative quantification was performed via the comparative $2^{-\Delta\Delta CT}$ method ⁷¹.

RNAi and RNAi efficiency determination

For lab-synthesized dsRNA, *gfp*, *NlCrzR* and *NlCrz* were amplified by PCR using specific primers conjugated with the T7 RNA polymerase promoter (primers listed in Table1). dsRNA was synthesized by the MEGAscript T7 transcription kit (Ambion, Austin, TX, USA) according to the manufacturer's instructions. Finally, the quality and size of the dsRNA products were verified by 1% agarose gel electrophoresis and the Nanodrop 1000 spectrophotometer and kept at -70°C until use.

3-5 days old virgin female and virgin male BPHs were used for injection of 50 nl / 30 nl of 5 µg/µl dsRNA per insect. Injection of an equal volume of *dsgfp* was used as negative control. After 48 hours, the insects were used behavior assays and gene relative expression analysis.

Neuropeptide injection after RNAi

48 hours after the *dsCrzR* injection, each insect was injected again with 10 ng of peptide dissolved in 50 nL of 1 × PBS balanced salt solution. Recovery was for six hours after injection. These insects were used for behavior assays. RNAi efficiency was examined by qPCR using a pool of ten individuals 48 hours after dsRNA injections.

In vitro synthesis of sgRNA and Cas9 mRNA

The single guide RNA (sgRNA) was designed as previously reported ⁷². Briefly, sgRNA was designed by manually searching genomic sequence around the region of the *Crz* and *CrzR* exon for the sequences corresponding to 5'-N₁₇₋₂₀NGG-3', where NGG is the protospacer-adjacent motif (PAM) of SpCas9 and N is any nucleotide. For in vitro transcription of sgRNA, were synthesized in vitro using the GeneArt™ Precision gRNA Synthesis Kit (Thermo Fisher Scientific, Vilnius, Lithuania) according to the instructions of manufacturer. The Cas9 protein (TrueCut™ Cas9 Protein v2, Cat. NO. A36497) was purchased from Thermo Fisher Scientific (Shanghai, China).

Embryo microinjection, crossing and genotyping

The embryonic injection was performed as previously reported ⁷³. The female BPHs, which had been mated for 3-6 days after emergence, were selected to lay eggs on fresh rice leaves. After 30 minutes, the embryos were transferred onto double-sided tape attached to a microscope slide along the long edge by gently pressing the slide onto the dorsal surface of embryos. A few drops of halocarbon oil 700 was added to these eggs. About 0.5–1 nL mixture of Cas9 protein (200 ng/µL) and sgRNA (100 ng/µL) was injected into each egg using a FemtoJet and Inject Man NI 2 microinjection system (Eppendorf, Germany). After the injection, the eggs were placed in Petri dishes covered with moist filter paper, and the Petri dishes were placed in plant growth chamber at 27±1 °C, 70±10% RH for hatching. These eggs hatched into G0 generations.

Crossing and genotyping (Re-edit)

To establish homozygous mutants of *Nilaparvata lugens* through CRISPR-mediated gene editing, newly emerged G0 adults were outcrossed with wild-type counterparts to generate G1 progeny. Following successful mating and oviposition, genomic DNA was extracted from G0 adults for mutation screening. The target region surrounding *NilCrzR* was amplified by PCR using specific primers *NilCrzR-check-F* and *NilCrzR-check-R* (primer sequences listed in [Table 1](#)), followed by Sanger sequencing to identify G0 individuals carrying mutations.

Progeny (G1) from mutation-positive G0 parents were maintained through sibling crosses to establish G2 populations. Subsequent genotyping via Sanger sequencing was performed on G1 individuals to select breeding pairs where both male and female parents carried heterozygous mutations. The resulting G2 offspring from these validated pairs were then subjected to intercrossing to ultimately generate homozygous mutant lines.

Quantitative RT-PCR

The first-strand cDNA was synthesized with HiScript II Q RT SuperMix for qPCR (+gDNA wiper) kit (Vazyme, Nanjing, China) using an oligo (dT)18 primer and 500 ng total RNA template in a 10 μ l reaction, following the instructions. Real-time qPCRs in the various samples used the UltraSYBR Mixture (with ROX) Kit (CWBI, Beijing, China). The PCR was performed in 20 μ l reaction including 4 μ l of 10-fold diluted cDNA, 1 μ l of each primer (10 μ M), 10 μ l 2 $\times$ UltraSYBR Mixture, and 6 μ l RNase-free water. The PCR conditions used were as follows: initial incubation at 95°C for 10 min, followed by 40 cycles of 95°C for 10 s and 60°C for 45 s. *N. lugens* 18S rRNA or *Drosophila* rp49 were used as an internal control ([Table 1](#)). Relative quantification was performed via the comparative $2^{-\Delta\Delta CT}$ method (Livak, K. J., 2001).

Immunohistochemistry

Adult BPHs, 3-5 days old, were dissected under 1x phosphate-buffered saline (PBS; pH 7.4) in Schneider's insect medium (S2). We dissected the brain, ventral nerve cord and reproductive system of the BPHs. These tissues were fixed in 4% paraformaldehyde in PBS for 30 min at room temperature. After extensive washing with PTX (0.5% Triton X100, 0.5% bovine serum albumin in PBS), blocked in 3% normal goat serum. Then, the tissues were incubated in Anti - CRZ for 12 hours at 4°C and in secondary antibody for 12 h at 4°C. Primary antibodies used were: rabbit anti - CRZ (1:1000, Anti - CRZ was gift from Jan A. Veenstra), mouse anti-Bruchpilot (1:1000, Developmental Studies Hybridoma Bank nc82). Secondary antibodies used: donkey anti-rabbit IgG conjugated to Alexa 488 (1:500, R37118, Thermo Fisher Scientific) and donkey anti-mouse IgG conjugated to Alexa 555 (1:500, R37115, Thermo Fisher Scientific). The samples were mounted in Vectorshield (Vector Laboratory). Images were acquired with Zeiss LSM 700 confocal microscopes, and were processed with Image J software. All antibodies were diluted in PTX solution. The primary antibody was diluted at a ratio of 1:1000, while the secondary antibody was diluted at a ratio of 1:500.

Adult female *Drosophila melanogaster*, 3-5 days old, were dissected under 1 x phosphate-buffered saline (PBS; pH 7.4) in Schneider's insect medium (S2). We dissected the brain, ventral nerve cord and reproductive system. These tissues were fixed in 4% paraformaldehyde in PBS for 30 min at room temperature. After extensive washing with PTX (0.5% Triton X100, 0.5% bovine serum albumin in PBS), blocked in 3% normal goat serum. Then, the tissues were incubated in Anti - CRZ and Anti - nc82 for 12 hours at 4°C and in secondary antibody for 12 h at 4°C. Primary antibodies used were: rabbit anti-CRZ (1:1000, A11122, Provided by Jan A. Veenstra). Secondary antibodies used: donkey anti-rabbit IgG conjugated to Alexa 488 and anti-mouse IgG conjugated to Alexa 555

Primer name	Sequence (5'-3')	Purpose
NICrz-F	ACCGCATCAGGACGACAATA	Gene clone
NICrz-R	ACCATAGCTGTTGAGCTACAAAT	
NICrz-RNAi-F	TAATACGACTCACTATAGGGGCTAG TGTTGCAATGCTGTTG	<i>dsNICrz</i> synthesis
NICrz-RNAi-R	TAATACGACTCACTATAGGGGCTGTT GAGCGTTAGACTGT	
QNICrz-F	AGACAGTGACCACAGCTCAA	qRT-PCR for NICrz
QNICrz-R	GAGTACTGGAACGTCTGTGC	
NICrz-sgRNA-F1	TAATACGACTCACTATAGGCACAGG AGTGATGCAGAC	sgRNA synthesis for NICrz knock-out
NICrz-sgRNA-R1	TTCTAGCTCTAAAACGTCTGCATCAC TCCTGTGCC	
NICrz-sgRNA-F2	TAATACGACTCACTATAGCTACTAGT GCTGGGCTGCA	sgRNA synthesis for NICrz knock-out
NICrz-sgRNA-R2	TTCTAGCTCTAAAACGTGAGCCAG CACTAGTAGC	
NICrz Check F	ACCATAGCTGTTGAGCTACAAAT	To analyze NICrz mutant
NICrz Check R	TGCAATGCTGTTGTGCGGAC	
NICrzR-F	AATCAGGCGATGAACCTCAGTT	Gene clone
NICrzR-R	ACGCAGTTCTATGACGTGAGG	
NICrzR-RNAi-F	TAATACGACTCACTATAGGGCGCCG TCTACACACTCATCT	<i>dsNICrzR</i> synthesis
NICrzR-RNAi-R	TAATACGACTCACTATAGGGCTACCA GCTTCGTACAGCGT	
QNICrzR-F1	CTGGTGCTAATAGGGCTGGA	qRT-PCR for NICrzR
QNICrzR-R1	TCCGGTGACAGTTGCTCTT	
QNICrzR-F2	TGCTCACCATCGCCTCTATC	qRT-PCR for NICrzR
QNICrzR-R2	TGCCATGGTTCGTGTAGAA	
NICrzR-F1	CCCACTAGCCAGCAACCATT	RT-PCR for NICrzR
NICrzR-R1	TCCTCCACAAATGGTCCACG	
NICrzR-sgRNA-F1	TAATACGACTCACTATAGGCACGCA CTCGAGCTGCCT	sgRNA synthesis for NICrzR knock-out
NICrzR-sgRNA-R1	TTCTAGCTCTAAAACAGGCAGCTCG AGTGCGTGCC	
NICrzR-sgRNA-F2	TAATACGACTCACTATAGGCGATGG TGAGCAGGTTGC	sgRNA synthesis for NICrzR knock-out
NICrzR-sgRNA-R2	TTCTAGCTCTAAAACGCAACCTGCTC ACCATCGCC	
NICrzR Check-F	GGCAGCAACCTGAGCCACCC	To analyze NICrzR mutant
NICrzR Check-R	GTCTTACCAGCGACCCAAGC	

Table 1.

Primers used in this study

QNlvG-F	GTGGCTCGTTCAAGGTTATGG	qRT-PCR for Nlvtellogenin
QNlvG-R	GCAATCTCTGGGTGCTGTTG	
QNlvGR-F	AGGCAGCCACACAGATAACCGC	qRT-PCR for Nlvtellogenin receptor
QNlvGR-R	AGCCGCTCGCTCCAGAACATT	
QNI53903-R	GTCCACCGAACTTTTCCTGG	
QNI Macc-F	TCTTTGGAAGTGGTGATTCCATCT	qRT-PCR for NIMacc
QNI macc-R	TGTCTACTTCACATACATGTGGCT	
QNI itpl-1-F	GATCAGATTCAATTACATTCTC	qRT-PCR for Nlitpl-1
QNI itpl-1-R	AGATTGTAGCAGTCCTCGCATA	

Table 1. (continued)

(R37118, Thermo Fisher Scientific). The samples were mounted in Vectorshield (Vector Laboratory). Images were acquired with Zeiss LSM 700 confocal microscopes, and were processed with Image J software. All antibodies were diluted in PTX solution.

Fluorescent in situ hybridization (FISH)

FISH was performed as previously reported by Yan et al. ⁷⁴. Briefly, the reproductive system of the female BPH dissection was performed under 1x Phosphate Buffered Saline (PBS) supplemented with a protease and phosphatase inhibitor cocktail (Shyuan, #R40012) on a clean, clear rubber pad using two fine forceps (Dumont, #0108-5-po). Dissected tissues were fixed using 4% paraformaldehyde (PFA) (Shyuan, #22039) diluted in 1 x PBS with 0.1% Tween20 (0.1% PBST) at room temperature (RT) for 20 min on a shaker. Fixed tissues were washed three times with 0.1% PBST for 15 min each on a shaker. The tissues were dehydrated in 25%, 50%, 90% and 100% methanol, and then rehydrated in reverse order, performed at room temperature for 10 minutes at a time. The tissue was then washed twice with 0.1% PBST and shaken for 5 minutes. A post fixation was performed by incubating the tissue in 4% PFA in 1 x PBS on a shaker for 20 minutes, followed by two washes with PBST for 5 minutes each while shaking. The tissue was permeabilized with proteinase K (10 µg/ml) at room temperature for 2 minutes. After incubation, the tissue was washed with 0.1% PBST for 5 minutes while shaking. The PBST was removed, and pre-hybridization solution (Boster, #AR0152) was applied at 55°C for approximately 2 hours. The DIG probes were denatured in a metal bath at 80°C for 5 minutes and then immediately placed on ice. The denatured DIG-labeled probes were added to the hybridization solution, and hybridization was carried out overnight at 55°C. Then, the hybridized tissue was washed twice at 55°C for 40 minutes with warm pre-hybridization solution, followed by four washes at room temperature with 0.1% PBST for 10 minutes each. The detection of the DIG-labeled probes was performed using an anti-DIG conjugated fluorescent antibody (1:100 dilution, Jackson, #200-542-156), incubated at room temperature in 0.1% PBST for 2 hours, followed by three washes with 0.1% PBST. Finally, the tissue was mounted in mounting medium and imaged using a Zeiss confocal microscope.

Imaging of reproductive organs

Male BPHs of different genotypes and different treatments, and female BPHs mated with males of different genotypes, were used for dissecting reproductive organs on the fourth day after emergence. Motic Images Devices and Motic Images Plus 3.0 were used to capture images of these reproductive organs.

Quantification and statistical analysis

All graphs were generated using Prism 9 software (GraphPad Software, La Jolla, CA). Data presented in this study were first verified for normal distribution by D'Agostino–Pearson normality test. If normally distributed, Student's *t* test was used for pairwise comparisons, and one-way ANOVA or two-way ANOVA was used for comparisons among multiple groups. If not normally distributed, Mann–Whitney test was used for pairwise comparisons, and Kruskal–Wallis test was used for comparisons among multiple groups, followed by Dunn's multiple comparisons. All data are presented as mean ± s.e.m.

Data availability

All of the RNA sequence data in this article have been deposited in the China National Center for Bioinformation database and are accessible in PRJCA045433. All data generated or analyzed during this study are included in the manuscript and supporting files; source data files have been provided for all figures.

Acknowledgements

We thank Dr. Jan A. Veenstra for providing CRZ antibody. This project was supported by National Key R&D Program of China (2022YFD1700200) to S.-F.W., the National Natural Science Foundation of China (No. 32022011 & 32472542) to S.-F.W..

Additional files

Figure 1 - Supplementary Video 1 

Additional information

Funding

National Key R&D Program of China (2022YFD1700200)

National Natural Science Foundation of China (32022011)

National Natural Science Foundation of China (32472542)

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Reviewer #1 (Public review):

In this work, Zhang et al, through a series of well-designed experiments, present a comprehensive study exploring the roles of the neuropeptide Corazonin (CRZ) and its receptor in controlling the female post-mating response (PMR) in the brown planthopper (BPH) *Nilaparvata lugens* and *Drosophila melanogaster*. Through a series of behavioural assays, micro-injections, gene knockdowns, Crispr/Cas gene editing, and immunostaining, the authors show that both CRZ and CrzR play a vital role in the female post-mating response, with impaired expression of either leading to quicker female remating and reduced ovulation in BPH. Notably, the authors find that this signaling is entirely endogenous in BPH females, with immunostaining of male accessory glands (MAGs) showing no evidence of CRZ expression. Further, the authors demonstrate that while CRZ is not expressed in the MAGs, BPH males with Crz knocked out show transcriptional dysregulation of several seminal fluid proteins and functionally link this dysregulation to an impaired PMR in BPH. In relation, the authors also find that in CrzR mutants, the injection of neither MAG extracts nor maccessin peptide triggered the PMR in BPH females. Finally, the authors extend this study to *D. melanogaster*, albeit on a more limited scale, and show that CRZ plays a vital role in maintaining PMR in *D. melanogaster* females with impaired CRZ signaling, once again leading to quicker female remating and reduced ovulation. The authors must be commended for their expansive set of complementary experiments. The manuscript is also generally well written. Given the seemingly conserved nature of CRZ, this work is a significant addition to

the literature, opening several avenues for testing the molecular and neurobiological mechanisms in which CRZ triggers the PMR.

However, there are some broad concerns/comments I had with this manuscript. The authors provide clear evidence that CRZ signaling plays a major role in the PMR of *D. melanogaster*, however, they provide no evidence that CRZ signaling is endogenous, as they did not check for expression in the MAGs of *D. melanogaster* males. Additionally, while the authors show that manipulating Crz in males leads to dysregulated seminal fluid expression and impaired PMR in BPH, the authors also find that CRZ injection in males in and of itself impairs PMR in BPH. The authors do not really address what this seemingly contradictory result could mean. While a lot of the figures have replicate numbers, the authors do not factor in replicate as an effect into their models, which they ideally should do.

Finally, while the discussion is generally well-written, it lacks a broader conclusion about the wider implications of this study and what future work building on this could look like.

<https://doi.org/10.7554/eLife.109297.1.sa1>

Reviewer #2 (Public review):

Summary:

The work presented by Zhang and coauthors in this manuscript presents the study of the neuropeptide corazonin in modulating the post-mating response of the brown planthopper, with further validation in *Drosophila melanogaster*. To obtain their results, the authors used several different techniques that orthogonally demonstrate the involvement of corazonin signalling in regulating the female post-mating response in these species.

They first injected synthetic corazonin peptide into female brown planthoppers, showing altered mating receptivity in virgin females and a higher number of eggs laid after mating. The role of corazonin in controlling these post-mating traits has been further validated by knocking down the expression of the corazonin gene by RNA interference and through CRISPR-Cas9 mutagenesis of the gene. Further proof of the importance of corazonin signalling in regulating the female post-mating response has been achieved by knocking down the expression or mutagenizing the gene coding for the corazonin receptor.

Similar results have been obtained in the fruit fly *Drosophila melanogaster*, suggesting that corazonin signalling is involved in controlling the female post-mating response in multiple insect species.

Notably, the authors also show that corazonin controls gene expression in the male accessory glands and that disruption of this pathway in males compromises their ability to elicit normal post-mating responses in their mates.

Strengths:

The study of the signalling pathways controlling the female post-mating response in insects other than *Drosophila* is scarce, and this limits the ability of biologists to draw conclusions about the evolution of the post-mating response in female insects. This is particularly relevant in the context of understanding how sexual conflict might work at the molecular and genetic levels, and how, ultimately, speciation might occur at this level. Furthermore, the study of the post-mating response could have practical implications, as it can lead to the development of control techniques, such as sterilization agents.

The study, therefore, expands the knowledge of one of the signalling pathways that control the female post-mating response, the corazonin neuropeptide. This pathway is involved in

controlling the post-mating response in both *Nilaparvata lugens* (the brown planthopper) and *Drosophila melanogaster*, suggesting its involvement in multiple insect species.

The study uses multiple molecular approaches to convincingly demonstrate that corazonin controls the female post-mating response.

Weaknesses:

The data supporting the main claims of the manuscript are solid and convincing. The statistical analysis of some of the data might be improved, particularly by tailoring the analysis to the type of data that has been collected.

In the case of the corazonin effect in females, all the data are coherent; in the case of CRISPR-Cas9-induced mutagenesis, the analysis of the behavioural trait in heterozygotes might have helped in understanding the haplosufficiency of the gene and would have further proved the authors' point.

Less consistency was achieved in males (Figure 5): the authors show that injection of CRZ and RNAi of *crz*, or mutant *crz*, has the same effect on male fitness. However, the CRZ injection should activate the pathway, and *crz* RNAi and mutant *crz* should inhibit the pathway, yet they have the same effect. A comment about this discrepancy would have improved the clarity of the manuscript, pointing to new points that need to be clarified and opening new scientific discussion.

<https://doi.org/10.7554/eLife.109297.1.sa0>